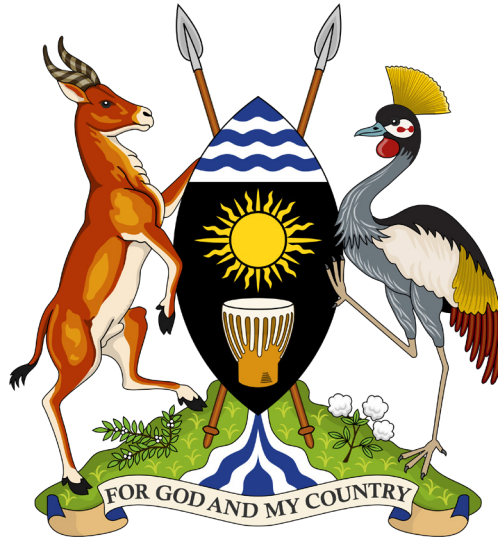




REPUBLIC OF UGANDA
MINISTRY OF HEALTH

The background of the cover features a series of vertical bars of varying heights, colored in shades of grey and yellow, on the left side. A large, bold, black diagonal line cuts across the middle of the page. On the right side, there are large, curved, overlapping shapes in yellow and red. The bottom of the page is a solid light yellow color.

SURVEILLANCE PLAN FOR ANTIMICROBIAL CONSUMPTION AND USE IN HUMAN HEALTH

JUNE 2025

ANTIMICROBIAL CONSUMPTION AND USE SURVEILLANCE PLAN IN HUMAN HEALTH

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FOREWORD

The Health Sector Development Plan (HSDP 2020/21-2024/25) and National Pharmaceutical Strategic Plan (NPSSP 2020/21 - 2024/25) demonstrate the Ministry of Health's commitment to addressing communicable diseases in Uganda. The Ministry has focused on the top three contributors to morbidity, mortality, and health care costs: HIV-AIDS, malaria, and lower respiratory tract infections. However, Antimicrobial Resistance threatens progress by reducing treatment options, increasing healthcare costs, and potentially raising mortality rates.

The joint ministerial collaboration under the One Health Platform developed the Antimicrobial Resistance National Action Plan (AMR NAP): a set of strategies for preventing and controlling the emergence and spread of resistant pathogens, including surveillance of antimicrobial consumption and use and ensuring antimicrobial stewardship.

The HSDP, NPSSP, and AMR NAP identify overuse and misuse of antimicrobials as the major modifiable driver of AMR and stress the need to preserve antimicrobial effectiveness through optimal access and appropriate use. This objective requires systematic collection and analysis of data on antimicrobial consumption and use from national to facility levels. The information will guide stewardship interventions and measure their impact against AMR.

This surveillance plan outlines a multipronged approach to generating information on antimicrobial consumption and use within the human health sector. It provides guidance on data management, including collection, analysis, and reporting, emphasizing the extent of antimicrobial overuse and misuse.

This AMCU surveillance plan sets priorities to optimally generate information guiding the attainment of health sector and national goals outlined in the HSDP 2021-2025 and AMR NAP 2024-2029.

I commend those who participated in developing this plan and thank all stakeholders who provided input under the coordination of the Department of Pharmaceuticals and Natural Medicines. I encourage all health facilities and partners in private and public sectors to implement this plan.

A handwritten signature in black ink, appearing to read 'Olaro', written in a cursive style.

Dr Charles Olaro
DIRECTOR GENERAL HEALTH SERVICES
MINISTRY OF HEALTH



ACKNOWLEDGEMENTS

The Ministry of Health has developed the National Surveillance Plan for Antimicrobial Consumption and Use (AMCU) in Human Health with funding from the Fleming Fund Country Grant. The development of this Antimicrobial Consumption and Use Surveillance Plan was spearheaded by the Makerere University Pharmacy Department (MakPD). This effort was conducted under the leadership of the Infectious Disease Institute (IDI) Limited, the consortium lead for Uganda's Fleming Fund Country Grant I.

We hereby thank the UK Department of Health and Social Care, the Fleming Fund, and Mott MacDonald Limited for their financial support in developing this plan.

The Department of Pharmaceuticals and Natural Medicines, the Ministry of Health, and the One-Health platform of the Government of Uganda gratefully acknowledge the technical assistance, guidance, and constructive comments provided by all stakeholders during the development process.

We also extend sincere gratitude to all other government agencies, such as the National Drug Authority, National Medical Stores, the local governments, and the regional referral and general hospitals, who cooperated extensively to ensure the comprehensive development of this AMCU surveillance plan and ensure that the final document is in line with national and international policies.

A handwritten signature in blue ink, appearing to be 'MGA', with a stylized flourish.

MARTHA GRACE AJULONG
Ag. COMMISSIONER HEALTH SERVICES: PHARMACEUTICALS
AND NATURAL MEDICINES

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ABBREVIATIONS

AMC	Antimicrobial consumption
AMCU	Antimicrobial Consumption and Use
AMEG	Antimicrobial Expert Group
AMR	Antimicrobial Resistance
AMU	Antimicrobial Use
ATC	Anatomical Therapeutic Chemical
CC	Collaborating Centre
DDD	Defined Daily Dose
DID	Defined Daily Doses/1000 inhabitants/day
DIS	Drug Indicator Survey
EMA	European Medicines Agency
EphMRA	European Pharmaceutical Market Research Association
GAP	Global Action Plan against AMR
GDP	Gross Domestic Product
GPPS	Global Point Prevalence Survey
INN	International Nonproprietary Name
INRUD	International Network for Rational Drug Use
IP	In-Patient Department
JMS	Joint Medical Stores
M&E	Monitoring and Evaluation
MAAIF	Ministry of Agriculture Animal Industry and Fisheries
MAUL	Medical Access Uganda Limited
MoH	Ministry of Health Uganda
MPM TWG	Medicines Procurement and Management Technical Working Group
MTC	Medicine and Therapeutics Committee
NAMRSC	National Antimicrobial Resistance Subcommittee
NAP	National Action Plan against AMR
NDA	National Drug Authority
NDAMIS	National Drug Authority Management Information System
NMS	National Medical Stores
OPD	Outpatient Department
OTC	Over-the-counter
PDD	Prescribed Daily Dose
PFP	Private-for-Profit
PIP	Pharmaceutical Information Portal
PIY	Packages/1000 inhabitants/year
PNFP	Private-not for-Profit
UHMG	Uganda Health Marketing Group
VC	Verification Certificate
WHA	World Health Assembly
WHO	World Health Organization

OPERATIONAL DEFINITIONS

Admissions	Sum of the patients admitted in a defined period.
Antimicrobial Resistance	A phenomenon in which microorganisms fail to respond to antimicrobial treatment that they were previously susceptible to.
Antimicrobial Agent	A naturally occurring, semi-synthetic or synthetic substance that exhibits antimicrobial activity (kill or inhibit the growth of microorganisms) at concentrations attainable in vivo. Anthelmintics and substances classed as disinfectants or antiseptics are excluded from this definition.
Antimicrobial Consumption data	refers to estimates of volumes of antimicrobials derived from aggregated data sources such as import, local manufacturer, distributor, wholesaler, and retailer, where there is no information available on the individual patients given the medicines or why the antimicrobials are being used. The assumption at this level is that all that is imported is consumed locally, and all that is distributed or sold is consumed locally, thus serving as a proxy estimate of the use of antimicrobials. Consumption data may be presented as total consumption for a country or disaggregated by setting (community or hospital; public or private sectors).
Antimicrobial Use data	refers to estimates derived from individual patient-level data. These data may allow disaggregation of data based on patient characteristics (age, gender, signs & symptoms, indication or diagnosis for which the medicine is being used), or route of administration and type of use (to treat, control or prevent infectious disease). Depending on the source of information, it may be possible to determine the medications ordered. This can facilitate the assessment of clinical practice against agreed protocols and standard treatment guidelines.
Discharges	Sum of the patients discharged in a defined period.
Granular data	is defined as data encompassing patient level use but also includes usage measured as volumes moving through the final level providers (e.g. hospitals, retailers including private pharmacies and drug shops etc.).
Health System	all organizations, people and actions whose primary intent is to promote, restore or maintain health.
Medicine Use Evaluation	involves assessing the use of a certain medication according to an established set of criteria. Criteria may relate to prescription (indication, dosages, frequency etc) or even administration/dispensing criteria (adherence to administration schedule, correct preparation and administration procedure etc).

Occupied bed days	Sum of occupied bed days in a defined period (number of bed days minus number of bed days without occupation by a patient).
Patient days	Sum of patient days for a defined period (e.g., for the year 2024 or 1st quarter of 2024).
A prescription audit	is a similar process, but the focus is to assess if a certain disease is treated according to set standard guidelines. It can be considered a partial “clinical audit”, which also involves a much wider assessment including structures, processes, competencies, skills, and outcomes in managing certain conditions.
Sales data	For surveillance data collection, sales data refers to the “amount (volumes) of antimicrobial agent(s) sold within the country for use in humans”. Sales data can be used as an approximation of actual use.
Surveillance	Ongoing, systematic collection, analysis and interpretation of health-related data essential to the planning, implementation, and evaluation of public health practice, closely integrated with the timely dissemination of these data to those responsible for prevention and control.
Usage data	is defined as estimates derived from patient level data and may include disaggregation to the level of specific patient characteristics (age, gender, or diagnosis).

CHAPTER 1

INTRODUCTION

1.1 The burden of AMR

Antimicrobial resistance (AMR) is a major threat to global health, as it makes common infections such as pneumonia, urinary tract infections, and gonorrhoea more difficult and expensive to treat due to resistance to the commonly available antimicrobial agents, leading to high mortality and morbidity rates. Resistance to antimalarials, antiretrovirals, and anti-tuberculosis agents will lead to declines in the progress made in reducing the burden of those killer diseases (WHO, 2017). In addition, the increased economic burden of AMR will be predominantly felt in low and middle-income countries (LMIC) due to the high cost of treating resistant infections, the need for longer hospital admissions, additional laboratory tests and expensive antibiotics, coupled with inadequate infrastructure and limited human and financial resources to deal with AMR. AMR, if left uncontrolled, is projected to lead to about 10 million deaths by 2050, the majority occurring in Africa and Asia (O'Neill, 2014).

Antimicrobial resistance is a naturally occurring phenomenon, but the way antimicrobials are used in the human health, animal health, and agricultural sectors is the strongest driver that accelerates the emergence and spread of AMR (Holmes, 2016). Irrational use of antimicrobials is common in the country, with antibiotics prescribed for up to 70% of patients in outpatient settings, including for non-bacterial infections (MOH Uganda, 2018). Reports have already shown the presence of resistance to common antimicrobials such as cotrimoxazole, ampicillin, and ceftriaxone, along with a high prevalence of multi-drug-resistant bacteria such as methicillin-resistant *Staphylococcus aureus* and extended-spectrum beta-lactamase (ESBL)-producers (UNAS, 2015).

In response to the growing threat of AMR, the World Health Organization (WHO) member states adopted the Global Action Plan (GAP) on AMR in 2015. In line with the global resolutions, Uganda launched a five-year national action plan (AMR NAP) to tackle AMR. The current NAP 2024-2029 outlines the actions to be taken to fight AMR through five strategic objectives that are: 1) promote public awareness, training and education, 2) improve infection prevention and control, 3) optimize access to and use of antimicrobials in human and animals, 4) surveillance and 5) invest in research and innovation.

1.2 Surveillance of Antimicrobial Consumption and Use

Antimicrobial use is the major modifiable factor contributing to AMR. The WHO GAP and Uganda AMR NAP consider antimicrobial stewardship, optimal access, and use as one of the strategies to prevent the emergence and spread of AMR. The goal is to preserve the effectiveness and efficacy of antimicrobial agents for human and animal health through controlled access, effective antimicrobial stewardship, and appropriate use. Stewardship interventions improve patient

outcomes and safety and reduce resistance so that patients get the best treatment they need while at the same time preserving the therapeutic effectiveness of antimicrobial agents.

1.2.1 The problem

The AMR NAP emphasizes the need for a robust, coordinated, and complementary system to routinely monitor the consumption and use of antimicrobials to measure existing practices, identify, prioritize, and investigate problems, and monitor the effects of interventions. The surveillance of antimicrobial consumption and use (AMCU) needs to be done in a systematic and standardised manner to produce good quality data that can be used to guide targeted interventions at both national and health facility levels.

However, Uganda currently lacks a structured and standardized surveillance system to monitor antimicrobial consumption and use (AMCU) at the national and health facility levels. While existing programs track the use of specific medicines like antimalarials, anti-tuberculosis, and HIV/AIDS commodities, no such system exists for antibiotics. Without reliable data on how and where antimicrobials are used, it is difficult to identify inappropriate prescribing practices, assess trends over time, and implement effective antimicrobial stewardship interventions, which undermines the efforts to combat AMR.

Addressing this requires the development of a system of surveillance that integrates data collection, analysis, and reporting on AMU in human health. This surveillance plan aims to establish a standard framework for monitoring antimicrobial usage and consumption, providing the necessary data to inform policy and practice, and contributing to the global effort to combat AMR.

1.2.2 Justification

Effective surveillance of antimicrobial consumption and use is essential for combating AMR and ensuring the rational use of antimicrobials. The World Health Organization (WHO) and Uganda's National Action Plan on AMR emphasize antimicrobial stewardship as a core strategy for preserving antimicrobial efficacy. Surveillance provides the necessary data to monitor antimicrobial prescribing and consumption trends, identify areas for intervention, and evaluate the impact of stewardship programs. The adoption of WHO's standardized metrics, such as Defined Daily Dose (DDD) per 1,000 inhabitants and the Access, Watch, and Reserve (AWaRe) classification, enables cross-country comparisons and benchmarking of antimicrobial use. Establishing a structured AMCU surveillance system in Uganda will support evidence-based decision-making at both national and facility levels, enhance policy implementation, and promote targeted stewardship efforts.

Systems thinking in monitoring AMCU

Systems thinking is an approach that views events of AMCU as part of a wider, dynamic system and involves much more than a reaction to events highlighted through surveillance. It requires a deeper understanding of the linkages, relationships, interactions, and behaviours among the elements that characterize the entire health system and influence AMCU. AMC and AMU are influenced by multiple factors, including prescribing behaviour, supply chain dynamics, patient demand, and regulations. Considering a systems approach to surveillance of AMCU in Uganda will support a comprehensive understanding of these interconnected factors, facilitating the development of effective, sustainable interventions.

This document, therefore, describes the national strategy for conducting AMCU surveillance in human health at both the national and subnational levels. The plan integrates the methods proposed in the WHO guidance, existing national policies and programmes on promoting appropriate medicines use in general, as well innovative approaches tailored to the Ugandan context.

The scope of surveillance detailed in this plan is limited to the broad aggregate consumption data from imports, local manufacture, and distribution, as well as the more granular patient data from health facilities.

CHAPTER 2

THE UGANDA NATIONAL PROGRAM ON ANTIMICROBIAL CONSUMPTION AND USE SURVEILLANCE IN HUMAN HEALTH

2.1 Aims and Objectives of AMCU Surveillance

2.1.1 Aim

The overall aim of antimicrobial use and consumption surveillance program is to generate evidence needed to identify trends in antimicrobial consumption and use. This evidence will guide clinical decisions regarding appropriate treatment and guide policy using the One Health approach.

A common protocol will be used for standardized collection, analysis and reporting of consumption and use data at national and subnational levels.

2.1.2 Objectives

The primary objectives of surveillance of AMCU are to:

- Provide reliable and comparable information on consumption and use of antimicrobials in the country.
- Promote the implementation of antimicrobial stewardship in the country.

2.2 Use of AMCU Surveillance Data

- Data on the consumption and use of antimicrobials have several uses, including to:
- Relate exposure to antimicrobials to the development of antimicrobial resistance.
- Identify and provide early warning signs of problems relating to changes in exposure to antimicrobials and develop interventions to address problems identified.
- Assess quality of prescribing and dispensing against practice guidelines.
- Raise Awareness of health professionals, consumers, and policy-makers about antimicrobial resistance and the contribution of inappropriate use of antimicrobials in humans.
- Detect trends and perform intra- and inter-facility comparisons.
- Plan, focus, monitor, and evaluate interventions aimed at changing exposure to antimicrobials.
- Support antimicrobial stewardship activities and provide the basis for setting targets for improving antimicrobial use.
- Monitor and evaluate the impact of pharmaceutical promotion on antimicrobial use

2.3 Governance and Operational Framework for the National AMCU Surveillance Program in Human Health

2.3.1 National Coordination

At the national level, oversight and overall coordination of the national AMCU surveillance will be provided by the Ministry of Health. The secretariat will be at the Department of Pharmaceuticals and Natural Medicines.

The Department of Pharmaceuticals and Natural Medicines will be responsible for:

1. Defining the objectives of the surveillance program.
2. Identifying, analyzing the data to be used in the surveillance program.
3. Developing guidelines and protocols for data collection to ensure good quality data
4. Reporting to inform the national strategy and publish the data at the national level.
5. Communicating and organizing meetings with the data providers to inform them of the purpose of the surveillance program and the requested data.
6. Provide technical support to sub-national teams and train health workers on how to carry out the AMCU surveys
7. Developing data quality assurance policies to be implemented at both national and subnational levels.
8. Develop recommendations to inform stewardship interventions on the use of antimicrobial medicines in the country.
9. Promoting the use of surveillance data to identify and monitor areas of misuse and overuse of antibiotics and to monitor and evaluate intervention programs against the set indicators.
10. Disseminating findings to the national level, data providers, and other AMR stakeholders.
11. Developing systems to support electronic data integration across sectors.
12. Securing resources and capacity to collect the data and to interpret them for use in public policy.
13. Monitoring and evaluating the implementation of the strategy and feed into the general M&E plan of the NAP.
14. Continuously reviewing the surveillance program to incorporate lessons learnt from previous activities to improve timeliness, efficiency and quality of AMCU data.
15. Liaising with other AMR surveillance teams in the animal health, laboratory, agricultural, and environmental sectors to implement the NAP.
16. Providing the report to the WHO through the global AMCU surveillance program

2.3.2 Health Facility AMCU

At the health facility level, the Medicines and Therapeutic Committee (MTC), through the Antimicrobial Stewardship (AMS) sub-committee, is responsible for performing AMCU activities. The AMS sub-committee is composed of clinicians, pharmacists, nurses, midwives, laboratory and administrative staff, and a representative from the IPC committee. The goal is to ensure that safe, effective, and cost-effective antimicrobials are made available to the facility and are used appropriately by both health facility staff and patients.

The duties of the AMS subcommittee in AMCU:

- On behalf of the MTC, lead on all aspects of antimicrobial use and misuse and provide feedback to all staff at the health facility.
- Assist in evaluating and selecting antimicrobials for the formulary and standard treatment guidelines.
- Develop policies concerning the use of antimicrobials within the health facility, including clauses that limit and restrict the use of antimicrobials in the hospitals and primary care clinics.
- Carry out antimicrobial consumption and use surveys and disseminate results to the health facility staff.
- Draft reports and forward them to the national level at the Ministry of Health.
- Monitor and assess consumption and use through prescribing quality assurance programs and medicine use evaluations to ensure the use of antimicrobials only when clinically indicated, in the correct dose, route, and for the appropriate duration.
- Participate in the educational programs for health-care staff.
- Collaborate with the Infection Prevention and Control (IPC) committee and laboratory departments to monitor and prevent/limit the emergence and spread of resistant microorganisms.

2.4 Stakeholders

Figure 1 depicts the stakeholders involved in AMCU surveillance and the flow of information between the different actors.

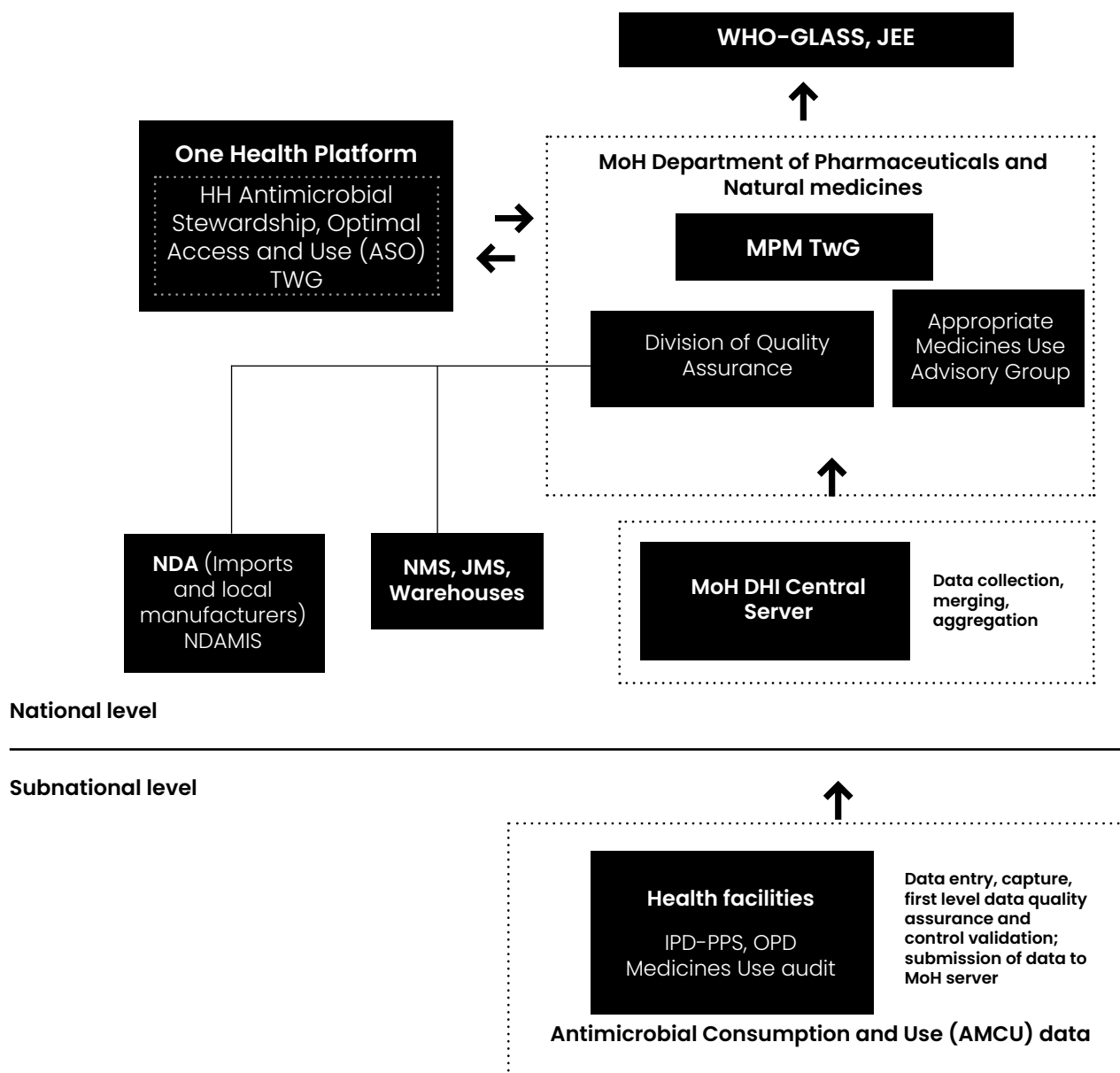


Figure 1: Schematic of stakeholders involved in AMCU Surveillance

CHAPTER 3

ANTIMICROBIAL CONSUMPTION AND USE SURVEILLANCE METHODS IN HUMAN HEALTH

3.1 Surveillance Systems

Surveillance is a critical tool for informing policies, guiding infection prevention and control measures, and assessing the spread of antimicrobial resistance. It plays a key role in monitoring the impact of local, national, and global strategies to optimize antimicrobial consumption and use (AMCU).

An effective surveillance system must align with the national objectives, strategies, and administrative structure and encompass both the public and private sectors to provide a comprehensive picture of AMCU trends. In Uganda, AMCU surveillance is integrated into the national surveillance framework and operates at two levels, as described in Figure 2: the national level and the health facility level. National-level surveillance focuses on tracking antimicrobial imports, locally manufactured drugs, and distribution data from central warehouses. At the health facility level, surveillance captures data on antimicrobial consumption and use within individual healthcare facilities.

Uganda's AMCU surveillance system not only informs national policies but also contributes to global AMR surveillance efforts. By generating reliable data on antimicrobial use and consumption, Uganda supports international initiatives such as the WHO Global Antimicrobial Resistance and Use Surveillance System (GLASS). These efforts enhance the global understanding of AMCU patterns, facilitate benchmarking across countries, and strengthen coordinated responses to combat AMR at regional and international levels.

3.1.1 Metrics for monitoring AMC

A key feature of surveillance of AMC is a measurable outcome, such as a reduction in antibiotic use to measure the success of antimicrobial stewardship interventions. Therefore, monitoring AMC relies on standardized metrics to ensure consistency, comparability, and meaningful interpretation of data. These standard units facilitate trend analysis and benchmarking. Key indicators include Defined Daily Dose (DDD) per 1,000 inhabitants per day, which estimates population-level antimicrobial exposure, and DDD per 100 bed-days, which assesses antimicrobial consumption in healthcare facilities. Additional metrics include the proportion of patients receiving antimicrobials, the quantity of antimicrobials procured or dispensed, and adherence to treatment guidelines.

Annex 1 and 2 provide details of standard antimicrobial utilization metrics and formulas for calculation.

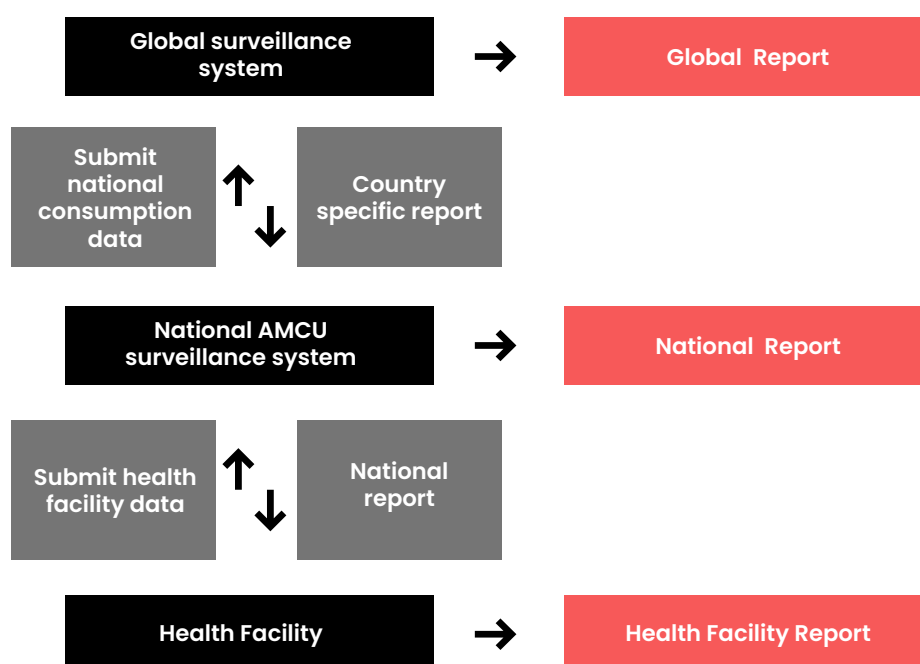


Figure 2: Data flow for surveillance at the hospital, subnational/national, and supranational level

3.2 Antimicrobials included in the surveillance

The monitoring of antimicrobial consumption is confined to antimicrobials intended for systemic use. A list of the respective antimicrobial groups is presented in Table 1.

Table 1: List of antimicrobial groups

Antimicrobial group	ATC-Code
Antibacterials for systemic use	J01
Antimycotics for systemic use	J02
Antimycobacterial	J04
Antivirals for systemic use	J05
Intestinal anti-infectives	A07A
Antifungals for systemic use	D01B
Antiprotozoals	P01

3.3 Information needed to calculate consumption for a given period

There are three elements to the data collection; namely, antimicrobial consumption data (numerator), denominator data and descriptive or contextual information that is relevant for interpreting the consumption estimates calculated.

Table 2 Information to calculate consumption

Element	Variables
Antimicrobial consumption data Product level data	For each antimicrobial included in the consumption analysis, the following variables are required: - The product (i.e as it appears when packaged) - Unique identifier of the medicine product package - Medicinal product package label (Brand Name) - WHO ATC category 5th Level Code (WHO latest version) - WHO ATC category 3rd Level Code (WHO latest version) - DDD per antibiotic as assigned by WHO-CC latest version - International Non-proprietary Name (INN) - Route of administration (Oral, parental, inhalation) - Is it a pediatric medicine product - Pharmaceutical formulation type - Strength of the active ingredient (e.g. tablet) - Unit of strength of the active ingredient (gm/mg/iu/mu) - Strength of volume and unit, if syrup - Package size of the product (e.g., 10 tablets) - Pack size unit of measure - Basic ingredient quantity and unit - Salt of the active substance - Total packages of the product consumed in a given period
Denominator data Used to calculate consumption density	Population under surveillance to which the data applies, and this may be stratified by health care sectors
Contextual information related to antimicrobial consumption	Data source information Which antimicrobials are included in surveillance? Specific exclusions, e.g. psychiatric facilities, private sectors

3.4 National Level AMC Surveillance

3.4.1. Imports and Local Manufacture Data

Import and local manufacture data will provide aggregated volume estimates of antimicrobial consumption in Uganda. This consumption data is useful for comparisons between Uganda and other countries or regions, evaluating trends of use over time, and assessing the impact of high-level policy. Surveillance at this level aims to quantify the extent of exposure to antimicrobials at the population level with a broad assumption that all that is imported or locally manufactured is consumed within the country and, therefore, serves as a proxy for use in the community. However, such high-level data cannot be used to assess accurately how antibiotics are used or where to target interventions.

Times of surveillance

Consumption surveillance should be done for each calendar year in line with the WHO guidelines.

Data sources

The source of data will be the National Drug Authority (NDA), which regulates and keeps records of all pharmaceutical products imported into and manufactured locally in the country (excluding any exports of products). The data will be obtained from the electronic records in the NDA Management Information System (NDAMIS) and customs records at NDA. A table listing the different data sources and their strengths and weaknesses is provided in Annex 2. Population estimates should be obtained from the Uganda Bureau of Statistics for the period under analysis.

Measurement of consumption

Antimicrobial consumption will be measured using aggregate data of all antimicrobials consumed in each period using a common system of nomenclature and standardized methodology. The Anatomical Therapeutic Chemical (ATC) classification system, and the number of Defined Daily Doses (DDDs) measurement metric will be used to measure consumption.

The Anatomical Therapeutic Chemical (ATC) system categorizes medicinal substances into different groups according to the organ or system on which they act and therapeutic, pharmacological and chemical properties. Each medicine is classified at five (5) different levels as in the table 3. More information on the ATC system is in Annex 3.

Table 3 Example of ATC coding

LEVEL	Example	
1st level, anatomical main group Consists of 1 letter	JA	Antiinfectives for systemic use Alimentary tract and metabolism
2nd level, pharmacological/ therapeutic subgroup	J01 J02 J04	Antibacterials for systemic use Antimycotics Antimycobacterials
3rd level, chemical/pharmacological sub group	J01C	Beta-lactam antibacterials, penicillins
4th level, chemical subgroup	J01CA	Penicillins with extended spectrum
5th level chemical substance	J01CA04 J01CA01	Amoxicillin Ampicillin

Unit of measurement – Defined Daily Dose (DDD)

The DDD is a measure used to quantify antimicrobial consumption and is assigned to each medicinal substance, taking into account the route of administration. DDD stands for the average maintenance dose per day for a medicine used for its main indication in adults. It should be considered that the DDD is a purely technical unit, which serves as a standard measure for quantification, but which does not necessarily reflect the recommended or factual use of the substance in the individual patient. The DDD is assigned for drugs that have an ATC code by the WHO Collaborating Centre for Drug Statistics Methodology, Norway. A list of antimicrobials for human use with their assigned DDDs (latest version) is provided in Annex 4 and accessed at http://www.whocc.no/atc_ddd_index/. Note: Always refer to the latest version for assigned DDDs, as they vary from time to time.

Consumption volume

The consumption volume will be expressed as the number of DDDs consumed and will be calculated by dividing the amount of the antimicrobial substance measured in grams by the DDD-value in grams, which has been assigned to the respective antimicrobial substance by the WHO Collaborating Centre for Drug Statistics Methodology.

The number of DDDs will be calculated as follows:

Number of DDD

$$= \frac{\text{(Total number of grams of the substance consumed in a defined period of time)}}{\text{(DDD value of the substance in grams assigned by WHO)}}$$

The total amount in grams can be obtained in two ways:

- by multiplying the strength of each tablet or vial by the number of items per package and the number of packages consumed; or
- by multiplying the strength of each tablet or vial by the total number of items consumed.

Examples for the calculation of the consumption volume are given in Annex 4. The DDD value is usually specified in grams but can also be defined as milligrams and million units (MU) for certain substances.

The total number of DDDs will then be aggregated by the desired ATC code Level for reporting purposes.

Consumption density

For standardization purposes, the volume of antimicrobial consumption (number of DDD, numerator) should be adjusted by measures representing the population (denominator) to get the consumption density.

Consumption density

$$= \frac{\text{(Number of DDD for time period P multiplied by 1000)}}{\text{(Quantity (number) of hospital activity indicator for time period P)}}$$

Examples for the calculation of the consumption density are given in Annex 2. Measures of hospital activity and hospital consumption data MUST correspond strictly with respect to the time period of surveillance and hospital units covered (e.g., whole hospital or single wards).

An Excel worksheet and guidance on steps for entering data calculating on antimicrobial consumption, volumes, and density using population can be found in WHO guidance on surveillance of antimicrobial consumption in hospitals; V0.1-20190619.

Analysis by AWaRe Categories

The antimicrobials consumed should also be analyzed by the AWaRe classification. To improve access and at the same time preserve the effectiveness of existing antibiotics, particularly those needed for second-line and “last resort” care, WHO has categorized antibiotics into three categories as AWaRe – Access, Watch and Reserve antibiotics. The Access category includes antibiotics needed for common infections and should be widely available and accessible in all countries. The Watch category include broad-spectrum antibiotics that should be used with caution because of their high potential to develop AMR, while the Reserve category is restricted as “last resort” antibiotics for multi-drug-resistant infections. Details are provided in Annex 5.

Reporting consumption estimates

The antimicrobial consumption will be reported as follows:

- total DDD/1000 inhabitants per day for the year X
- DDDs disaggregated by ATC Level 3 Code, which is the antimicrobial chemical or pharmacological group for the year X
- DDDs stratified by ATC Code and AWaRe Classification for the year X
- Proportions of antimicrobials in the UCG or Essential Medicines List
- Discussion of results “in context”

3.4.2 Surveillance at National Centralized Distributors

Centralized level warehouse distribution records can also provide aggregated data of antimicrobial consumption that act as a proxy for use to the facilities or institutions which they serve. The consumption can then be disaggregated or stratified by region/geographical location, by health facility level of care, by public or private sectors, and again by population level metrics using the population estimates of the regions supplied.

The main centralized level warehouses distributing antimicrobials nationwide include:

- National Medical Stores, supplies to all public health facilities, Joint Medical Stores, supplies to PNFP and private health facilities
- Efforts will be made to include other private warehouses and distributors with national and regional scope.

The methodology and data collection tools used at this level are similar to those used for imports and local manufacture AMC surveys described in the sections above.

Volume data on antimicrobials distributed will be expressed as the total DDDs per 1000 inhabitants. The denominator for consumption data will be the population of the geographical location to which the medicines have been distributed. The data may also be disaggregated by antibiotic class, level of care, geographical location, and the WHO AWaRe classification

3.5 Surveillance at the health facility level

Facility level data provides a more accurate granular assessment on how antimicrobials are actually used. Client specific data can provide information on volume and type of antibiotics dispensed across provider types (public, and private facilities), and end user types. AMU data at the granular level can be used for surveillance, target setting, monitoring interventions or behavior change, communication of risks, and comparison at the same level of care.

3.5.1 Routine reports on antimicrobial use at facility level

Supervision, Performance Assessment and Recognition Strategy (SPARS) : aims to increase health workers' ability to manage medicines through on-the-job training and support from Medicines Management Supervisors (MMS).

Data from SPARS over time have described a trend of inappropriate medicine use.

3.5.2 Other activities for AMCU surveillance at the health facility

- Antimicrobial consumption surveys using WHO ATC/DDD methodology.
- WHO medicine indicator surveys for outpatient department assessment of Use.
- Point Prevalence surveys for in-patient use assessments.

- Prescription audits and medicine use evaluations for selected antimicrobials.
- Medicines cost studies for selected antimicrobials.

3.5.3 Facility antimicrobial consumption surveys

Consumption data at the health facility is described as the quantity of a particular antimicrobial issued from the main store for a defined period of time. The data may be aggregated at the store level to represent the total consumption for the facility or disaggregated by ward. Data can be collected at, in-patient or outpatient, ward level, departmental level, specialty level and whole hospital level.

Antimicrobials monitored

These include all antimicrobials described in section 3.2, but specifically antimicrobials in the ATC group J01 and antimicrobials for systemic use must be included in the surveillance.

Variables collected for the antimicrobial product

These are similar to those in section 3.4. In addition, the following information must be collected.

- Description of the facility including name of facility and the level of care
- Period under consideration: month, quarter and year.
- Unit of issue for each individual drug
- Description of the antimicrobial that is ATC classification and AWaRe classification.
- Total volume issued out of the store within the period of consideration, as DDD.
- Volumes issued out of the store disaggregated by user department, as DDD

Data collection

The data collection is based on the WHO Methodology for AMC surveillance in hospitals (version 2019). It uses the ATC/DDD methodology to calculate consumption, and an Excel template is available. Instructions for filling the Excel template are provided in WHO guidance on surveillance of antimicrobial consumption in hospitals; V0.1-20190619.

Variables for denominator data to measure consumption density (in-patient department)

In order to calculate the consumption density, the volume of the antimicrobials (number of DDDs) is adjusted by measures representing the hospital activity indicators for the period of surveillance. A list of different hospital activity indicators, which can be used for normalizing the number of DDDs consumed, is presented in the table below. A detailed description of the different denominators, including examples for calculations, is provided in the table below.

Table 4 Variables to measure consumption density

Hospital activity measure	Description
Total Patient Days (TPD)	The sum of patient days for a defined period of time (e.g., for the year 2023 or 1st quarter of 2024)
Occupied bed days (OBD)	Sum of occupied bed days in a defined period of time (number of bed days minus number of bed days without occupation by a patient)
Admissions (ADM)	Sum of the patients admitted in a defined period of time
Discharges (DIS)	Sum of the patients discharged in a defined period of time

The consumption is then expressed as the number of DDDs per quantity of hospital activity (e.g., DDDs per 100 patient days or 100 admissions) for the specified time period and calculated as in the formula below.

$$\text{Consumption density} = \frac{(\text{Number of DDD for time period P multiplied by 1000})}{(\text{Quantity (number) of hospital activity indicator for time period P})}$$

Note: It must be considered that the use data of the hospital activity MUST correspond strictly to the consumption data. For example, if the consumption data have been collected for a whole year, the hospital activity data have to be collected for the same whole year.

Sources of data

Sources of the data at health facility level may be:

- Purchase/procurement data of the hospital administration or pharmacy. This may be obtained from electronic medical records such as (Clinic Master, eAFYA, and Rx Solution), Stock cards and stock books, Invoices from the drug suppliers.
- Unit-based dispensing data; data on antimicrobials dispensed to departments/wards by stores, e.g. from issue and requisition vouchers.
- Patient-based dispensing/billing data; data on antimicrobials sold or dispensed to individual patients e.g. from dispensing logs.
- Prescription data; data on prescription of antimicrobials e.g. OPD register, in-patient register and electronic health records.
- Patient registers to obtain the total number of patients in the ward at a given time or period of time

Data analysis and reporting

- The data will be analyzed as the number of DDDs consumed per hospital/department/ward, then as proportions of ATC Level 3 categories (antimicrobial class) and proportions by AWaRe categories.
- Data will be interpreted contextually, that is, knowledge about the wider framework of surveillance, e.g., hospital type, size, and structure.
- The used data sources and the coverage of hospital consumption data are helpful for sensible data interpretation.
- Monitoring of the respective contextual information is useful in order to document changes over time (such as in hospital structure, data sources used), which might facilitate the interpretation of longitudinal consumption data.

Period of surveillance

Data collection shall be performed at least once a year with at least yearly aggregated data. In order to validate the effect of intervention measures or for other antimicrobial stewardship purposes, it can be meaningful to gather and analyse data in shorter time intervals, e.g. quarterly or monthly.

3.5.4 Prescription and Medicine Use Audit/ Evaluation

Data on antimicrobial use at the OPD shall be collected using the Prescription and medicine use Audit/Evaluation tool in OPD, adapted from the Ministry of Health's Medicines and Therapeutics Committee manual and AMS manual.

The Prescription and medicine use Audit/Evaluation is a standardized method of assessing prescribing patterns at the facility. It answers the question of, “how are medicines being used in primary care practice?” and provides a broad overview of the prevalence of antibiotic use at OPD. The OPD antibiotic survey narrows down to specifics of the use of individual antibiotics. It answers the question, “for what diagnosis is an individual antibiotic being used?” or “what antibiotics are being used for a particular diagnosis?” The number, route, type of antibiotics, type of diagnosis, need of antibiotics (based on a recorded diagnosis of infectious disease or not), and adherence to treatment guidelines can be analyzed.

The data sources are usually the OPD register (HMIS 002), prescription forms, or electronic medical records. The methodology for conducting the Prescription and medicine use Audit/ Evaluation is provided in annexes 8, 9, and 10.

Indicators of interest for prescription and Medicine use Audit/Evaluation (sample at least 100 patients/prescriptions) include;

- Average number of medicines per prescription/per patient
- Percentage of patients receiving one or more antibiotics
- Percentage of patients receiving one or more injections
- Percentage of medicines prescribed by generic name
- Percentage of medicines that are antibiotics
- Percentage of medicines that are injections
- Percentage of antibiotics that are injections
- Percentage of medicines prescribed in the Essential Medicines and Health Supplies List (the latest version)
- Percentage of patients on antibiotics with a definitive diagnosis

Table 5: MoH prescribing indicators and reference standards for OPD

Prescribing indicators	MoH Standard*
Average number of medicines per encounter	Less or equal to 2.6
Percentage of encounter with antibiotics prescribed	Less or equal to 40%
Percentage of encounters with injection prescribed	Less or equal to 15%
Percentage of drugs prescribed by generic name	100%
Percentage of drugs from essential drug list	100%

***Adapted from WHO reference standards**

With regard to patients for whom antibiotics are prescribed, the indicators of interest are:

- The most common diagnoses for which antibiotics are prescribed
- Most commonly prescribed antibiotics
- Diagnoses for which the most common antibiotic is prescribed (medicine use evaluation)
- Most prescribed antibiotics for the most common condition at OPD at the time of the survey
- Percentage of prescriptions that are in accordance with Standard Treatment Guidelines (UCG/ Institutional guidelines)

- Number of antibiotics per prescription

The results of the prescription and medicine use evaluation can be used to guide on which antibiotics need further analysis. For example, the most common antibiotic used or the most common diagnosis can then be subjected to a prescription audit or medicine use Audit.

3.5.5 Point Prevalence Surveys

Point prevalence surveys are used to assess the use of antibiotics in in-patients. The parameters of interest are collected for all the patients admitted in a certain ward at a certain moment.

Different standardized surveys have been developed, but the Ministry of Health has adapted the WHO Methodology for Point Prevalence Survey on Antibiotic Use in Hospitals, version 1.1 (available at https://www.who.int/medicines/access/antimicrobial_resistance/WHO-EMP-IAU-2018_01/en/, accessed on December 5th 2019) as the main surveillance tool to assess antibiotic use among hospitalized patients.

There are other methods such as the Global PPS of Consumption and Resistance (University of Antwerp, Biomeraux-fund), PPS of Healthcare Associated Infection and Antimicrobial Use (ECDC), and British Society of Antimicrobial Chemotherapy (BSAC).

The general purpose of PPS is to provide a standardized mechanism to be used by hospitals in inpatient departments to assess prescription practices, identify areas for quality improvement, and assess the effectiveness of Antimicrobial Stewardship interventions. The Global PPS also aims to establish a global surveillance tool through voluntary internet-based reporting.

The PPS normally includes all patients present at 8 am in the morning in the ward under consideration and having received antibiotics in the previous 24 hours. Parameters assessed are usually:

- Antimicrobial name (generic and branded)
- Route of administration (parenteral, oral, rectal),
- Indication of antimicrobial use (community-acquired infection, Hospital-Acquired Infection, Surgical prophylaxis, Medical prophylaxis)
- Diagnosis (if present in notes, and site and type of infection)
- Presence of a stop/review date
- Type of treatment: empiric vs targeted.
- Is treatment in compliance with standard treatment guidelines
- Type of biological fluid sample: blood, urine, etc.
- Is the treatment based on a biomarker? Which one?
- Which group of resistant organisms was found?

A simplified data collection tool is presented below. Links to more information and material are provided in the references provided in Annex 10

Data source

The source of data for PPS is the in-patient records.

Data analysis

The PPS results can be summarised and reported as:

- The most common diagnoses for which antibiotics are prescribed
- Most commonly prescribed antibiotics
- Diagnoses for which the most common antibiotic is prescribed (medicine use evaluation)
- Most prescribed antibiotics for the most common condition at that time (prescription audit)
- Percentage of prescriptions that are in accordance with the standard treatment guidelines (e.g., UCG)
- Percentage of patients receiving more than one antibiotic per prescription, disaggregated by the actual number of antibiotics prescribed.

CHAPTER 4

IMPLEMENTATION MATRIX

This comprehensive implementation matrix outlines the key activities required to activate the antimicrobial consumption and use (AMCU) surveillance plan. It translates strategic goals and objectives into actionable steps, detailing sub-activities, verification methods, timelines, locations, responsible parties, estimated costs, and funding sources. The matrix serves as a guide for stakeholders at both national and subnational levels to coordinate their efforts, monitor progress, and ensure accountability in executing the surveillance plan. It is structured around the plan's core objectives.

Objective 1: Provide reliable and comparable information on national consumption and use of antimicrobials in health facilities.				
Strategy/Activity	Means of Verification	Location	Responsible Entity	Source of funding
Strategy 1: Improve governance and coordination				
1.1: Carry out stakeholder engagement meetings (National):	Functional NCC in place with staff and TORs	National	MoH Commissioner Pharmacy	MOH
	Functional Appropriate Medicine Advisory committee with TORs	National	MoH Commissioner Pharmacy	MOH
	Orientation report	National	MoH Commissioner Pharmacy	MOH
	Meeting reports	National	MoH Commissioner Pharmacy	MOH/Partners
1.2: Carry out stakeholder engagement meetings (Local Government)	Meeting reports	Sub-National	Regional Pharmacists	MOH/Partners
Strategy 2 Strengthen capacity in surveillance through training, mentorship and equipment acquisition				
2.1: Disseminate, print, and distribute AMCU surveillance guidelines, plans, and SOPs	Dissemination workshop report, Copies of AMCU guidelines printed and distributed	National	Commissioner Pharmacy	MOH/Partners
2.2: Procure equipment for the Appropriate Medicines Use unit to support data management	Equipment procured and installed at the Appropriate Medicines Use unit	National	Commissioner Pharmacy	MOH/Partners
2.3: Train and routinely mentor staff at the Appropriate Medicines Use unit on data management.	Training reports	National	Commissioner Pharmacy	MOH/Partners
2.4: Train and routinely mentor stakeholders involved in AMCU surveillance on data management and reporting	Training reports	National Sub-National	Commissioner Pharmacy	MOH/Partners
2.5: Provide support supervision to stakeholders involved in AMCU surveillance.	Support supervision reports	Sub-National	Commissioner Pharmacy	MOH/Partners

Strategy 3: Conduct data collection on antimicrobial imports and local manufacture				
3.1 Conduct assessment of antimicrobial imports through the Integrated Regulatory Management Information systems (IRIMS)	One annual data set/report on antimicrobials imported	National	Commissioner Pharmacy, National Drug Authority (NDA)	NDA
3.2 Conduct assessment of antimicrobial local manufacturing through IRIMSS	Annual reports on antimicrobials locally manufactured	National	Commissioner Pharmacy, National Drug Authority (NDA)	NDA
Strategy 4: Conduct data collection on antimicrobial distribution to public and PNFP facilities				
4.1: Conduct assessments of volumes of antimicrobials distributed	Antimicrobial warehouse stock reports submitted to the appropriate medicines use unit	National	Commissioner Pharmacy NMS, JMS, MAUL	MOH/Partners
Strategy 5: Conduct data collection on antimicrobial consumption and use at facility level				
5.1: Survey of antimicrobial consumption data from the store: expenditure on antibiotics, ward consumption	Annual antimicrobial consumption data from the store	Sub-National	Commissioner Pharmacy, Facility MTC/AMS sub-committee	MOH/Partners
5.2: Point Prevalence survey for antimicrobial use in IPD and drug indicator surveys for antimicrobial use in OPD at RRH and NRH levels of care	PPS and DIS data sets received at the appropriate medicines use unit	Sub-National	Commissioner Pharmacy, Facility MTC/AMS sub-committee	MOH/Partners
5.3: Point Prevalence survey for antimicrobial use in IPD and drug indicator surveys for antimicrobial use in OPD at district hospitals and HCIV levels of care	Quarterly PPS and DIS data sets received at the appropriate medicines use unit	Sub-National	Commissioner Pharmacy, Facility MTC/AMS sub-committee/AMS Focal persons	MOH/Partners
5.4: Point Prevalence survey for antimicrobial use in IPD and drug indicator surveys for antimicrobial use in OPD from private hospitals	Quarterly PPS and DIS data sets received at the appropriate medicines use unit	Sub-National	Commissioner Pharmacy	MOH/Private sector/ Partners
Strategy 6: Conduct data entry and storage				
6.1: Safe and secure storage on a local network in accordance with national ethical and data safety regulations	A regularly backed up database for AMCU surveillance created in the National Health Information Database	National	Commissioner of Pharmacy, Appropriate Medicines Use Unit	MOH/Partners
6.2: Direct entry and storage using a dedicated web-based tool on a national host server.	Web-based tool developed	National	Commissioner Pharmacy, Appropriate Medicines Use Unit	MOH/Partners
Strategy 7: Conduct data analysis				
7.1 Appropriate Medicines Use Unit reviews and analyses aggregated data from different data providers	Analysed data sets, aggregate reports generated	National	Appropriate Medicines Use Unit, Health Facilities	MOH/Partners
7.2 Appropriate Medicines Use unit compiles annual reports and provides feedback to sites	Annual reports, minutes of review meetings	National	Appropriate Medicines Use Unit, Appropriate Medicines Use Unit / Advisory Group	MOH/Partners
Strategy 8: Data reporting				
8.1: Data reported to the Appropriate Medicines Use unit	Data sets received at Appropriate Medicines Use Unit, National Health Information System Database	National	Appropriate Medicines Use Unit, Health Facilities	MOH/Partners
8.2: Reports discussed by the Appropriate Medicines Use Advisory Group and the Medicines Procurement and Management TwG within the ministry to describe actionable areas	Reports received for discussion by Appropriate Medicines Use Advisory Group, Minutes from AMU advisory Group and MPM TWG meeting	National	Appropriate Medicines Use Unit; Appropriate Medicines Use Unit / Advisory Group	MOH/Partners
8.3: Reports shared with: The One Health Platform through the One Health Surveillance TwC and the antimicrobial stewardship, optimal access and use technical working group (ASO)	Reports received at the One Health secretariat through the Data Integration and sharing center (DISC). WHO-Glass receipt	National	Appropriate Medicines Use Unit, One Health Platform	MOH/Partners

Objective 2: promote implementation of antimicrobial stewardship in health facilities.				
Sub-Activity	Means of Verification	Location	Responsible Entity	Source of Funding
Strategy 9: Promote use of Surveillance Data				
9.1: Surveillance data used to develop and refine data quality indicators of AMU/C	Data quality indicators developed based on baseline data	National	Appropriate Medicines Use Unit/ Advisory Group	MOH
9.2: Data used to identify targets for antimicrobial use improvement at both national and facility level	Targets for Quality improvement set identified, minutes from facility MTC and MOH AMS subcommittee meetings	National Sub-national	Appropriate Medicines Use Unit/ Advisory Group	MOH
9.3: Data used to inform and design interventions (including policies) to improve antimicrobial use at the National level	Updated national policies and guidelines on antimicrobial use developed	National	Appropriate Medicines Use Unit/ Advisory Group	MOH
9.4: Data used to inform and design interventions to improve antimicrobial use at the Facility level	Facility work plans and other evidence of data use at the facility level	Sub-national	Facility MTC/AMS sub-committee	MOH
9.5: Data used to monitor the effectiveness of antimicrobial stewardship activities	Impact assessment reports	National Sub-national	Appropriate Medicines Use Unit/ Advisory Group Facility MTC/AMS sub-committee	MOH

CHAPTER 5

DETAILED COSTING OF SURVEILLANCE

Strategies/Objectives/Activities	2024	2025	2026	2027	2028	Total
Objective 1: Provide reliable and comparable information on national consumption and use of antimicrobials in health facilities						
Strategy 1: Improve Governance and Coordination	100,000,000	0	100,000,000	0	100,000,000	300,000,000
Activity 1.1: Carry out stakeholder engagement meetings (National)	40,000,000	0	40,000,000	0	40,000,000	120,000,000
Activity 1.2: Carry out stakeholder engagement meetings (Local government)	60,000,000	0	60,000,000	0	60,000,000	180,000,000
Strategy 2: Strengthen capacity in surveillance through training, mentorship and equipment acquisition	290,000,000	75,000,000	315,000,000	75,000,000	315,000,000	1,070,000,000
Activity 2.1: Disseminate, print, and distribute AMCU surveillance guidelines, plans, and SOPs	30,000,000	0	0	0	0	30,000,000
Activity 2.2: Procure equipment for the Appropriate Medicines Use unit to support data management	20,000,000	0	0	0	0	20,000,000
Activity 2.3: Train and routinely mentor staff at the Appropriate Medicines Use unit on data management	40,000,000	0	40,000,000		40,000,000	120,000,000
Activity 2.4: Train and routinely mentor stakeholders involved in AMCU surveillance on data management and reporting	200,000,000	0	200,000,000		200,000,000	600,000,000
Activity 2.5: Provide support supervision to stakeholders involved in AMCU surveillance	0	75,000,000	75,000,000	75,000,000	75,000,000	300,000,000
Strategy 3: Conduct data collection on antimicrobial imports and local manufacture	20,000,000	20,000,000	20,000,000	20,000,000	20,000,000	100,000,000
Activity 3.1: Conduct assessment of antimicrobial imports through the Integrated Regulatory Management Information systems (IRIMS)	10,000,000	10,000,000	10,000,000	10,000,000	10,000,000	50,000,000
Activity 3.2: Conduct assessment of antimicrobial local manufacturing through IRIMS	10,000,000	10,000,000	10,000,000	10,000,000	10,000,000	50,000,000

Strategy 4: Conduct data collection on antimicrobial distribution to public and PNFP facilities	10,000,000	10,000,000	10,000,000	10,000,000	10,000,000	50,000,000
Activity 4.1: Conduct assessments of volumes of antimicrobials distributed	10,000,000	10,000,000	10,000,000	10,000,000	10,000,000	50,000,000
Strategy 5: Conduct data collection on antimicrobial consumption and use at facility level	134,000,000	134,000,000	134,000,000	134,000,000	134,000,000	670,000,000
Activity 5.1: Survey of antimicrobial consumption data from the store: expenditure on antibiotics, ward consumption	20,000,000	20,000,000	20,000,000	20,000,000	20,000,000	100,000,000
Activity 5.2: Point Prevalence survey for antimicrobial use in IPD and drug indicator surveys for antimicrobial use in OPD at RRH and NRH levels of care	38,000,000	38,000,000	38,000,000	38,000,000	38,000,000	190,000,000
Activity 5.3: Point Prevalence survey for antimicrobial use in IPD and drug indicator surveys for antimicrobial use in OPD at district hospitals and HCIV levels of care	38,000,000	38,000,000	38,000,000	38,000,000	38,000,000	190,000,000
Activity 5.4: Point Prevalence survey for antimicrobial use in IPD and drug indicator surveys for antimicrobial use in OPD from private hospitals	38,000,000	38,000,000	38,000,000	38,000,000	38,000,000	190,000,000
Strategy 6: Conduct data entry and storage	30,000,000	0	0	0	0	30,000,000
Activity 6.1: Safe and secure storage on a local network in accordance with national ethical and data safety regulations	30,000,000	0	0	0	0	30,000,000
Activity 6.2: Direct entry and storage using a dedicated web-based tool on a national host server.	0	0	0	0	0	0
Strategy 7: Conduct data analysis	24,000,000	24,000,000	24,000,000	24,000,000	24,000,000	120,000,000
Activity 7.1: Review and analyses of aggregated data from different data providers	12,000,000	12,000,000	12,000,000	12,000,000	12,000,000	60,000,000
Activity 7.2: Compile annual reports and provide feedback to sites	12,000,000	12,000,000	12,000,000	12,000,000	12,000,000	60,000,000
Strategy 8: Data reporting	11,000,000	11,000,000	11,000,000	11,000,000	11,000,000	55,000,000
Activity 8.1: Discussion of Reports by the Appropriate Medicines Use Advisory Group and the Medicines Procurement and Management TwG within the ministry to describe actionable areas	3,000,000	3,000,000	3,000,000	3,000,000	3,000,000	15,000,000
Activity 8.2: Sharing of Reports with: The One Health Platform through the One Health Surveillance TwC and the antimicrobial stewardship, optimal access and use technical working group (ASO)	3,000,000	3,000,000	3,000,000	3,000,000	3,000,000	15,000,000

Activity 8.3: Sharing and dissemination of Reports with international development partners (WHO) and at scientific meetings.	5,000,000	5,000,000	5,000,000	5,000,000	5,000,000	25,000,000
Objective 2: Promote implementation of antimicrobial stewardship in health facilities						
Strategy 9: Promote use of Surveillance Data	15,000,000	15,000,000	15,000,000	15,000,000	15,000,000	75,000,000
Activity 9.1: Develop and refine data quality indicators of AMU/C using surveillance data	3,000,000	3,000,000	3,000,000	3,000,000	3,000,000	15,000,000
Activity 9.2: Identify targets for antimicrobial use improvement at both national and facility level based on AMC/U data	3,000,000	3,000,000	3,000,000	3,000,000	3,000,000	15,000,000
Activity 9.3: Use data to inform and design interventions (including policies) to improve antimicrobial use at the National level	3,000,000	3,000,000	3,000,000	3,000,000	3,000,000	15,000,000
Activity 9.4: Use data to inform and design interventions to improve antimicrobial use at the Facility level	3,000,000	3,000,000	3,000,000	3,000,000	3,000,000	15,000,000
Activity 9.5: Use data to monitor the effectiveness of antimicrobial stewardship activities	3,000,000	3,000,000	3,000,000	3,000,000	3,000,000	15,000,000
TOTAL	634,000,000	289,000,000	629,000,000	289,000,000	629,000,000	2,470,000,000

CHAPTER 6

MONITORING AND EVALUATION PLAN FOR AMCU IN HUMAN HEALTH

6.0 Monitoring and Evaluation

There is need for an M&E system that provides timely and accurate information to the Ministry of Health and its implementing partners in order to inform performance reviews, policy discussions and periodic revisions to the overall AMCU surveillance plan.

6.1 Goals and Objectives of M&E

6.1.1 Goal

The goal for M&E is to assess progress towards establishing a system that is robust, comprehensive, fully integrated, harmonized, and well-coordinated to generate reliable and comparable data to inform AMCU surveillance interventions.

6.1.2 Objectives

The specific objectives for M&E of the AMCU surveillance plan are:

- To provide a framework for tracking progress and demonstrating results of the AMCU Surveillance plan over the medium term.
- Capacity within relevant structures (such as the Appropriate Medicines Use Unit and the AMCU subcommittee) to systematically track progress in implementing the AMCU surveillance plan.
- To support the Appropriate Medicines Use Unit within the Ministry of Health's Pharmacy Department and other stakeholders in assessing performance according to agreed objectives and performance indicators, thereby facilitating evidence-based decision-making.
- To improve compliance with broader government policies, including the Pharmaceutical Sector Strategic Plan and the National Action Plan.
- To promote continuous learning by documenting and sharing challenges and lessons learned among stakeholders during the implementation of the surveillance plan.

6.2 Key M&E outcomes

The expected key outputs of the M&E framework are:

1. Performance reports such as baseline survey reports, periodic progress reports, annual performance reports, financial audit reports, among others .

2. Basic statistical data on health service delivery, resources, outputs, and beneficiaries. Collection of basic statistical data on health service delivery, resources, outputs, and beneficiaries.
3. Regular updates on core performance indicators.

The M&E Plan should result in:

- Timely reporting on progress of implementation of the AMCU surveillance plan;
- Timely meeting of reporting obligations to Ministry of Health;
- Objective decision making for performance improvement
- Accountability to the Ministry of Health
- Policy dialogue with stakeholders.
- Evidence-based policy development and advocacy.
- Institutional memory on AMCU surveillance implementation

	Indicator	Indicator Definition	Indicator Level	Disaggregation	Data Source	Frequency	Responsibility
1.0 Leadership, Organization, and Management							
1.1	Designation of the Human Health AMCU National Coordination Center (NCC)	Presence of a designated Human Health AMCU NCC	Output		AMCU surveillance assessments, Appointment letters or Terms of Reference	Annual	MoH/NAMRSC
1.2	a functional AMS subcommittee at the MoH- Pharmacy Department	Presence of a functional AMCU Sub-committee at the MoH- Pharmacy Department	Output		A signed minutes from meeting and attendance lists, Acceptance of appointment to the committee, Approved annual work plan	Quarterly	MoH/NAMRSC
1.3	Presence of a multi-disciplinary team coordinating AMCU at the facility	Presence of a multi-disciplinary team coordinating AMCU at the facility	Output		AMCU surveillance assessments Appointment letter for the different cadres on Facility AMCU coordinating team, TORs	Annual	MoH/NAMRSC
	Presence of a multi-disciplinary team coordinating AMCU at the facility	Existence of an annual activity work plan for the AMS Sub-Committee	Output		Approved work plan	Annual	MoH/AMS sub-Committee
2.0 Capacity for AMCU Surveillance							
2.1	Proportion of MTC health staff trained in AMCU surveillance	N: Number of health staff trained D: Total number of MTC members	Output	By data provider level	Training reports, MTC Reports	Biannual	AMS Sub-committee

2.2	Proportion of health facilities submitting AMCU surveillance reports to MoH	N: Number of facilities reporting AMCU surveillance data to MoH D: Expected number of facilities reporting AMCU surveillance data	Output	By facility level	DHIS2/National Health Data Warehouse (NHDW) We need to specify which DHIS2 instances receive this data	Biannual	AMS Sub-committee
2.3	Percentage of reports submitted on time to the MoH	N: Number of facilities reporting AMCU surveillance data to MoH on time D: Expected number of facilities reporting AMCU surveillance data	Output	By facility level	DHIS2/National Health Data Warehouse (NHDW) We need to specify which DHIS2 instances receive this data	Biannual	AMS Sub-committee
2.4	Number of AMCU review meetings conducted	N: Number of human health AMCU review meetings conducted	Process		Meeting minutes	Biannual	AMS Sub-committee
3.0 Antimicrobial consumption							
3.1	Volume of antimicrobial medicines imported	Total volumes of antimicrobials imported	Output	WHO AWaRe Classification	NDA IRIMS Integrated national quantification National annual needs analysis and quantification for public health facilities in Uganda	Annual	NDA/MoH/NMS
3.2	Volumes of antimicrobial medicines locally manufactured	To track volume of antimicrobials locally manufactured	Output	WHO AWaRe	NDAMIS	Annual	NDA
3.3	Proportion of antibiotics locally manufactured against country needs	N: Volume of antibiotics manufactured D: Country needs	Output	WHO AWaRe	NDAMIS Integrated national quantification	Annual	NDA
3.4	Volumes of antibiotics distributed to public and PNFP facilities	N: Volumes of antibiotics distributed in public and PNFP facilities	To track the volumes & classes of antimicrobials distributed to the different levels of care	Distribution reports	By Level of care, ATC classification, WHO AWaRe, WHO critically important antibiotics	Quarterly	Output
3.4	National Antibiotic Consumption data published and submitted to GLASS	N: National Antibiotic consumption reports published, National Antibiotics reports submitted to GLASS	Output	By Level of care	Ware house stock reports Delivery reports	Annual	National Warehouses, NDA (NMS, JMS), MOH DPNM
3.5	Percentage of financial expenditure on antibiotics	N: Total cost of all antimicrobials purchased D: Total cost of all medicines purchased	Outcome		Electronic logistics management information systems Delivery Notes, Order Forms	Bimonthly	Facility AMCU Team

3.6	Percentage of encounters with an antibiotic prescribed	N: Total number of prescriptions with antibiotics D: Total number of patients sampled	Outcome	By department (OPD, IPD)	OPD register, dispensing logs, inpatient files	Biannual	Facility AMCU Team
3.7	Average number of antibiotics prescribed per patient encounter	N: Total number of antibiotics prescribed for all patients sampled D: Total number of medicines prescribed for all patients sampled	Outcome	By department (OPD, IPD)	OPD register, dispensing logs, inpatient files	Biannual	Facility AMCU Team
4.0 Prescription Quality							
4.1	Percentage of antibiotics prescribed consistent with the institutional formulary list or UCG EMHSL	N: Number of antibiotics prescribed that are on the formulary list/ UCG EMHSL D: Total number of antibiotics prescribed	Outcome	By Department (OPD, IPD)	Inpatient files, OPD Register , institutional formulary list (EMHSL), UCG	Biaannual	Facility AMCU Team
4.2	Average duration of administered antibiotics Add for OPD	N: Total number of days of antibiotic administration D: Total number of hospital bed days at the point of the survey	Outcome	IPD	In patient files	Biannual	Facility AMCU Team
4.3	Percentage of patients with Upper-Respiratory Tract Infections (URTI) who were prescribed with antibiotics	N: Total number of patients diagnosed with URTI and antibiotics prescribed D: Total number of patients diagnosed with URTI	Outcome	OPD	OPD register	Biannual	Facility AMCU Team
4.4	Percentage of antibiotics prescribed by generic name	N: Total number antibiotics prescribed by generic name for sampled patients D: Total number antibiotics prescribed for the sampled patients	Outcome	By department (OPD, IPD)	OPD register, dispensing logs, In-patient files	Biannual	Facility AMCU Team
4.5	Percentage of encounters with an injectable antibiotic prescribed at OPD	N: Number of encounters for which an injectable antibiotic was prescribed at OPD D: Total number of encounters for which an antibiotic prescribed at OPD	Outcome	OPD	OPD register, dispensing logs,	Biannual	Facility AMCU Team

I ANNEXES

Annex 1: Metrics for monitoring antimicrobial consumption and use

Unit of Measurement	Definition	Advantages	Disadvantages
Antimicrobial Consumption			
Daily Defined Dose (DDD)	The assumed average maintenance dose per day for a drug used for its main indication in adults	The most used unit allows benchmarking, cross-sectional and longitudinal, and is independent of price, package size, and strength	DDD may change over the years, cannot be used in pediatrics or in patients that need special doses, and does not always reflect the doses used in real practice
Defined daily dose per 1000 inhabitants per day (DID)	Represents the volume of antimicrobials used at population level	Adjusted to the real situation of the hospital, department, or type of patient	The availability of population-level data often lags in time
Prescribed Daily Dose (PDD)	The average dose prescribed per day according to a representative sample of prescriptions of a hospital or ward. - PDD/*DDD > 1 may indicate over prescription or higher doses than recommended. - PDD/*DDD < 1 may indicate under dosing, which could lead to treatment failure or antimicrobial resistance (AMR). *Assigned DDD, not the total DDDs		Not standardized, only used at facility-level analysis, does not usually allow comparisons or benchmarking between hospitals and departments
Days of therapy (DOT)	Number of days that a patient receives a certain antibiotic, independently of the quantity and used doses	Not influenced by discrepancy between DDD and PDD, can be used in pediatrics, independent of DDD changes	Doses and intensity of the doses used are not considered, difficult to measure without a computerized system at the patient level
Days of Therapy per 1000 inhabitants/day (DOTID)	Represents the total number of days of antibiotic exposure at a population level	More accurate estimate of exposure than DDD/DID in pediatric populations and individuals requiring lower doses	Requires population-level AMU data, including information about the duration of therapy

Occupied bed-days	One occupied bed-day refers to one inpatient hospital bed occupied by a patient for one day, calculated from the time of admission to discharge	The most used unit	Does not allow a full interpretation of consumption trends
Admissions (or discharges)	Number of patients from outside hospital who are admitted to a hospital department and occupy beds	Shows activity changes in the hospital and helps to interpret the trends of consumption over years	Difficult to obtain separately for hospital departments
Antimicrobial Use			
Antimicrobial Prescribing Rate (APR)	A measure of antibiotic initiation relative to a population or sub-population, patient years or patient visits	APR may be a more clinically relevant metric to influence behaviour change in prescribers than measures of total antimicrobial use such as DID	Does not address appropriateness of antimicrobial selection or duration
Proportion of prescriptions containing an antimicrobial (Prevalence of antimicrobial use)	Number of antibiotic prescriptions divided by the total number of prescriptions	Can be determined using prescription information only	Subject to variability in overall prescribing volume which may not be an accurate reflection of patient volume
Proportion of prolonged antimicrobial course prescribed	Describes duration of therapy; number of antimicrobial prescriptions > specific number of days	(i) Optimal metric to evaluate quality of prescribing (ii) Prolonged durations represent an important component of inappropriate antibiotic prescribing	This does not address the antimicrobial selection

Annex 2: Calculation of AMCU Metrics

Unit of Measurement	Formula
Antimicrobial Consumption	
Total Daily Defined Dose (DDD)	(Total grams of the substance consumed in a defined period)/(DDD value of the substance in grams assigned by WHO)
Defined daily dose per 1000 inhabitants per day (DID)	(Total DDDs)/(Population x 365) x 1000
Prescribed Daily Dose (PDD)	(Total daily doses prescribed)/(Number of doses per day)
Days of therapy (DOT)	Sum of number of days a patient receives at least one dose of an antibiotic
Days of Therapy per 1000 inhabitants/day (DOTID)	(DOT of all antibiotics prescribed)/(Population x 365) x 1000
Occupied bed-days	Sum of the inpatient beds occupied per day X number of days
Admissions (or discharges)	Sum of patients admitted to a facility in a given period
Antimicrobial Use	
Antimicrobial Prescribing Rate (APR)	(Total antibiotic prescriptions)/Population x 1000
Proportion of prescriptions containing an antimicrobial (Prevalence of antimicrobial use)	(Antibiotic prescriptions)/(Total admissions) x 100
Proportion of prolonged antimicrobial course prescribed	(Antibiotic courses > 8 days)/(Total number of antibiotic courses) x 100

Calculation of total DDDs at import and distribution level

A. Calculation based on the number of packages

Number of DDD = ((number of packages) × (package size) × (strength per item))/(Assigned DDD)

Example:

Number of DDD = (20 tablets × 50 packages × 0.75g)/1.5g = 500 DDD

B. Calculation based on the number of items

Number of DDD = ((number of items) × (strength per single item))/(Assigned DDD)

Example:

Number of DDD = (1000 tablets × 0.75 g)/(1.5 g) = 5000 DDD

It is not necessary to provide data on the number and size of packages, if data on the number of items (e.g. tablets) is available.

Special cases

For combinations of antibiotics, the DDD value is specified as UD (Unit Dose). One tablet or vial of a combination product with a specific strength of each component is defined as a specific number of UD representing the DDD. To obtain the DDD consumed of a specific combination product, the total number of UDs is divided by the assigned UD value. A list of combination products with specified strengths and their assigned DDD values is provided by the WHO Collaborating Centre for Drug Statistics Methodology.

For combination products, which contain one active and one inactive ingredient the DDD is ordinarily assigned only to the active ingredient. For example, for amoxicillin/clavulanic acid the DDD is assigned to amoxicillin and neglects the clavulanic acid. In consequence, for the calculation of the number of DDD for amoxicillin/clavulanic acid, the total volume of grams consumed should refer only to amoxicillin as in the example below.

Examples for data entry of combined products

Example A – Data entry of combined products consisting of one active antimicrobial substance and one inactive substance

The substance amoxicillin/clavulanic acid includes as active antimicrobial substance amoxicillin and as an inactive additive clavulanic acid. For the quantification of antimicrobial consumption only the active substance is relevant, and the DDD-value assigned by the WHO collaboration center only refers to amoxicillin.

From the product «amoxicillin/clavulanic acid, 875mg/125mg, 10 tablets » 7 packages have been dispensed to the ward « PNEU1 » in the first quarter of 2018. The consumption volume in DDD is calculated by multiplying the number of the items dispensed (number of packages x number of items per package) with the strength per single item (only amoxicillin) divided by the DDD-value assigned by WHO (1.5 g).

Consumption volume = $((7 \text{ packages} \times 10 \text{ tablets}) \times 0.875 \text{ g}) / (1.5 \text{ g}) = 40.8 \text{ DDD}$

Table 6: Product-level data variables for the antimicrobial register

Variable Name	Content
Product ID	Unique identifier of the medicinal product package e.g. registration number
Label	Medicinal product package label (Brand Name)
Pack size unit	Pack size unit of measure
Paediatrics Product	Is it a pediatric medicine product
Form	Pharmaceutical formulation type
Route Admin.	Route of administration
Strength	Quantity of the main ingredient of each item
Strength Unit	Unit measurement of strength
INBASQ	Basic ingredient quantity
INBASQ_UNIT	Unit measurement of the basic ingredient quantity
ATC5	WHO ATC code at substance level (ATC-5 level)
SALT	Salt of the active substance
COMBINATION	The WHO CC has defined DDD for combined products
PRODUCT NAME	Medicinal product name
INGREDIENTS	Ingredient name: e.g. amoxicillin
PRODUCT_ORIGIN	The product could be an import, donation or locally manufactured
Manufacturer Country	The country of the manufacturer of the product
Manufacturer	Name of the manufacture
Generic	Is the product a generic?
Conv_Factor	Transform strength expressed in IU in grams
WHO_DDD	The DDD defined by the WHO CC for the ATC code
WHO_DDD_Unit	Unit measurement of the WHO DDD (mg, g, IU)
DPP	DDD per package

Annex 3: Data Sources for AMCU Data

There are a number of sources of information on consumption of antimicrobials, providing information with differing levels of detail on the consumption and use of antimicrobials, as discussed in the table below.

Table 7: Strengths and limitations of data sources for AMCU surveillance

Data source	Strengths	Limitations
Import data(NDAMIS)	- Centralized records - Standardized reporting for customs declaration forms including product type (generic, branded), volume, port of origin, country of manufacture, batch number, expiry date)	- Documentation may be incomplete - Does not account for parallel imports or illegal entry of products - Volumes match import cycles, not consumption patterns. A single Verification Certificate from NDA will allow for import of a product throughout the year - Administrative records are not formatted for research and analysis, therefore require further manipulation to generate meaningful information. - Currently, NDA/NDAMIS is using European Pharmaceutical Market Research Association (EphMRA) classification rather than Anatomical Therapeutic Chemical codes, so there may be limited information at the pharmacological or chemical subgroup level
Local manufacturer/distributor (NDAMIS)	- Local licensed producers can easily be identified by the system - Separate product volumes for local use and for export can be separated - Data can be requested in a format suitable for analysis that doesn't require further manipulation	- There is no policy requiring local manufacturers to file returns with NDA. - There may be unwillingness to provide data - Volumes reflect production not consumption patterns
Central-Warehouses (NMS, JMS, distribution data)	- Data is likely to have accurate documentation of purchases - Disaggregation of distribution data to facility types (community and hospital) and geographical location is possible - Few procurement agencies in this category - Donations imported for use at these facilities are channeled through these agencies, including programme commodities	- Only provides data for public sector and PNFPs - The data may not reflect total public sector consumption if other procurement is undertaken by hospitals, health facilities
Facility level: hospital pharmacy dispensing or procurement databases, or, when available, aggregated figures from electronic prescribing systems. Other sources, Stock cards, Stock books, issue/requisition vouchers)	- Standardized data collection - Capacity to combine data from multiple sources including manufacturer records (batch number expiry date), volumes received at the facility, demanders/ward volumes ordered	May be limited data collection in some areas, especially those relying on manual records May not be able to examine data at prescriber level
Patient level consumption data (Dispensing logs)	Data may be disaggregated by patient demographic characteristics	May be difficult to trace a patient to a prescription and to the dispensing log
Prescribing records (OPD Registers and Patient files for those in admission)	This data has patient characteristics, diagnosis, dose, duration, and other prescribed medicines	- Prescribed medicines may not be dispensed - Data in the registers and files may not be complete, and underestimate the prevalence of antibiotic use - Samples of patients may not be representative and therefore not reflect national data

Annex 4: List of antimicrobials with ATC codes and DDD values

ATC Codes	Antimicrobials	DDD values
J	Antiinfectives for systemic use	1
J01	Antibacterials for systemic use	2
J01A	Tetracyclines	3
J01AA	Tetracyclines	4
J01AA01	Demeclocycline	5
J01AA02	Doxycycline	5
J01AA03	Chlortetracycline	5
J01AA04	Lymecycline	5
J01AA05	Metacycline	5
J01AA06	Oxytetracycline	5
J01AA07	Tetracycline	5
J01AA08	Minocycline	5
J01AA09	Rolitetracycline	5
J01AA10	Penimepicycline	5
J01AA11	Clomocycline	5
J01AA12	Tigecycline	5
J01AA13	Eravacycline	5
J01AA20	Combinations Of Tetracyclines	5
J01AA56	Oxytetracycline, Combinations	5
J01B	Amphenicols	3
J01BA	Amphenicols	4
J01BA01	Chloramphenicol	5
J01BA02	Thiamphenicol	5
J01BA52	Thiamphenicol, Combinations	5
J01C	Beta-Lactam Antibacterials, Penicillins	3
J01CA	Penicillins With Extended Spectrum	4
J01CA01	Ampicillin	5
J01CA02	Pivampicillin	5
J01CA03	Carbenicillin	5
J01CA04	Amoxicillin	5
J01CA05	Carindacillin	5
J01CA06	Bacampicillin	5
J01CA07	Epicillin	5
J01CA08	Pivmecillinam	5
J01CA09	Azlocillin	5
J01CA10	Mezlocillin	5
J01CA11	Mecillinam	5
J01CA12	Piperacillin	5
J01CA13	Ticaracillin	5
J01CA14	Metampicillin	5
J01CA15	Talampicillin	5
J01CA16	Sulbenicillin	5
J01CA17	Temocillin	5

J01CA18	Hetacillin	5
J01CA19	Aspoxicillin	5
J01CA20	Combinations	5
J01CA51	Ampicillin, Combinations	5
J01CE	Beta-Lactamase Sensitive Penicillins	4
J01CE01	Benzylpenicillin	5
J01CE02	Phenoxymethylpenicillin	5
J01CE03	Propicillin	5
J01CE04	Azidocillin	5
J01CE05	Pheneticillin	5
J01CE06	Penamercillin	5
J01CE07	Clometocillin	5
J01CE08	Benzathine Benzylpenicillin	5
J01CE09	Procaine Benzylpenicillin	5
J01CE10	Benzathine Phenoxymethylpenicillin	5
J01CE30	Combinations	5
J01CF	Beta-Lactamase Resistant Penicillins	4
J01CF01	Dicloxacillin	5
J01CF02	Cloxacillin	5
J01CF03	Meticillin	5
J01CF04	Oxacillin	5
J01CF05	Flucloxacillin	5
J01CF06	Nafcillin	5
J01CG	Beta-Lactamase Inhibitors	4
J01CG01	Sulbactam	5
J01CG02	Tazobactam	5
J01CR	Combinations Of Penicillins, Incl. Beta-Lactamase Inhibitors	4
J01CR01	Ampicillin And Beta-Lactamase Inhibitor	5
J01CR02	Amoxicillin And Beta-Lactamase Inhibitor	5
J01CR03	Ticarcillin And Beta-Lactamase Inhibitor	5
J01CR04	Sultamicillin	5
J01CR05	Piperacillin And Beta-Lactamase Inhibitor	5
J01CR50	Combinations Of Penicillins	5
J01D	Other Beta-Lactam Antibacterials	3
J01DB	First-Generation Cephalosporins	4
J01DB01	Cefalexin	5
J01DB02	Cefaloridine	5
J01DB03	Cefalotin	5
J01DB04	Cefazolin	5
J01DB05	Cefadroxil	5
J01DB06	Cefazedone	5
J01DB07	Cefatrizine	5
J01DB08	Cefapirin	5
J01DB09	Cefradine	5
J01DB10	Cefacetrile	5

J01DB11	Cefroxadine	5
J01DB12	Ceftezole	5
J01DC	Second-Generation Cephalosporins	4
J01DC01	Cefoxitin	5
J01DC02	Cefuroxime	5
J01DC03	Cefamandole	5
J01DC04	Cefaclor	5
J01DC05	Cefotetan	5
J01DC06	Cefonicid	5
J01DC07	Cefotiam	5
J01DC08	Loracarbef	5
J01DC09	Cefmetazole	5
J01DC10	Cefprozil	5
J01DC11	Ceforanide	5
J01DC12	Cefminox	5
J01DC13	Cefbuperazone	5
J01DC14	Flomoxef	5
J01DD	Third-Generation Cephalosporins	4
J01DD01	Cefotaxime	5
J01DD02	Ceftazidime	5
J01DD03	Cefsulodin	5
J01DD04	Ceftriaxone	5
J01DD05	Cefmenoxime	5
J01DD06	Latamoxef	5
J01DD07	Ceftizoxime	5
J01DD08	Cefixime	5
J01DD09	Cefodizime	5
J01DD10	Cefetamet	5
J01DD11	Cefpiramide	5
J01DD12	Cefoperazone	5
J01DD13	Cefpodoxime	5
J01DD14	Ceftibuten	5
J01DD15	Cefdinir	5
J01DD16	Cefditoren	5
J01DD17	Cefcapene	5
J01DD18	Cefteram	5
J01DD51	Cefotaxime And Beta-Lactamase Inhibitor	5
J01DD52	Ceftazidime And Beta-Lactamase Inhibitor	5
J01DD54	Ceftriaxone, Combinations	5
J01DD62	Cefoperazone And Beta-Lactamase Inhibitor	5
J01DD63	Ceftriaxone And Beta-Lactamase Inhibitor	5
J01DD64	Cefpodoxime And Beta-Lactamase Inhibitor	5
J01DE	Fourth-Generation Cephalosporins	4
J01DE01	Cefepime	5
J01DE02	Cefpirome	5

J01DE03	Cefozopran	5
J01DF	Monobactams	4
J01DF01	Aztreonam	5
J01DF02	Carumonam	5
J01DH	Carbapenems	4
J01DH02	Meropenem	5
J01DH03	Ertapenem	5
J01DH04	Doripenem	5
J01DH05	Biapenem	5
J01DH06	Tebipenem Pivoxil	5
J01DH51	Imipenem And Cilastatin	5
J01DH52	Meropenem And Vaborbactam	5
J01DH55	Panipenem And Betamipron	5
J01DI	Other Cephalosporins And Penems	4
J01DI01	Ceftobiprole Medocaril	5
J01DI02	Ceftaroline Fosamil	5
J01DI03	Faropenem	5
J01DI54	Ceftolozane And Beta-Lactamase Inhibitor	5
J01E	Sulfonamides And Trimethoprim	3
J01EA	Trimethoprim And Derivatives	4
J01EA01	Trimethoprim	5
J01EA02	Brodinoprim	5
J01EA03	Iclaprim	5
J01EB	Short-Acting Sulfonamides	4
J01EB01	Sulfaisodimidine	5
J01EB02	Sulfamethizole	5
J01EB03	Sulfadimidine	5
J01EB04	Sulfapyridine	5
J01EB05	Sulfafurazole	5
J01EB06	Sulfanilamide	5
J01EB07	Sulfathiazole	5
J01EB08	Sulfathiourea	5
J01EB20	Combinations	5
J01EC	Intermediate-Acting Sulfonamides	4
J01EC01	Sulfamethoxazole	5
J01EC02	Sulfadiazine	5
J01EC03	Sulfamoxole	5
J01EC20	Combinations	5
J01ED	Long-Acting Sulfonamides	4
J01ED01	Sulfadimethoxine	5
J01ED02	Sulfalene	5
J01ED03	Sulfametomidine	5
J01ED04	Sulfametoxydiazine	5
J01ED05	Sulfamethoxypyridazine	5
J01ED06	Sulfaperin	5

J01ED07	Sulfamerazine	5
J01ED08	Sulfaphenazole	5
J01ED09	Sulfamazone	5
J01ED20	Combinations	5
J01EE	Combinations Of Sulfonamides And Trimethoprim, Incl. Derivatives	4
J01EE01	Sulfamethoxazole And Trimethoprim	5
J01EE02	Sulfadiazine And Trimethoprim	5
J01EE03	Sulfametrole And Trimethoprim	5
J01EE04	Sulfamoxole And Trimethoprim	5
J01EE05	Sulfadimidine And Trimethoprim	5
J01EE06	Sulfadiazine And Tetroxoprim	5
J01EE07	Sulfamerazine And Trimethoprim	5
J01F	Macrolides, Lincosamides And Streptogramins	3
J01FA	Macrolides	4
J01FA01	Erythromycin	5
J01FA02	Spiramycin	5
J01FA03	Midecamycin	5
J01FA05	Oleandomycin	5
J01FA06	Roxithromycin	5
J01FA07	Josamycin	5
J01FA08	Troleandomycin	5
J01FA09	Clarithromycin	5
J01FA10	Azithromycin	5
J01FA11	Miocamycin	5
J01FA12	Rokitamycin	5
J01FA13	Dirithromycin	5
J01FA14	Flurithromycin	5
J01FA15	Telithromycin	5
J01FA16	Solithromycin	5
J01FF	Lincosamides	4
J01FF01	Clindamycin	5
J01FF02	Lincomycin	5
J01FG	Streptogramins	4
J01FG01	Pristinamycin	5
J01FG02	Quinupristin/Dalfopristin	5
J01G	Aminoglycoside Antibacterials	3
J01GA	Streptomycins	4
J01GA01	Streptomycin	5
J01GA02	Streptoduocin	5
J01GB	Other Aminoglycosides	4
J01GB01	Tobramycin	5
J01GB03	Gentamicin	5
J01GB04	Kanamycin	5
J01GB05	Neomycin	5
J01GB06	Amikacin	5

J01GB07	Netilmicin	5
J01GB08	Sisomicin	5
J01GB09	Dibekacin	5
J01GB10	Ribostamycin	5
J01GB11	Isepamicin	5
J01GB12	Arbekacin	5
J01GB13	Bekanamycin	5
J01M	Quinolone Antibacterials	3
J01MA	Fluoroquinolones	4
J01MA01	Ofloxacin	5
J01MA02	Ciprofloxacin	5
J01MA03	Pefloxacin	5
J01MA04	Enoxacin	5
J01MA05	Temafloxacin	5
J01MA06	Norfloxacin	5
J01MA07	Lomefloxacin	5
J01MA08	Fleroxacin	5
J01MA09	Sparfloxacin	5
J01MA10	Rufloxacin	5
J01MA11	Grepafloxacin	5
J01MA12	Levofloxacin	5
J01MA13	Trovafloxacin	5
J01MA14	Moxifloxacin	5
J01MA15	Gemifloxacin	5
J01MA16	Gatifloxacin	5
J01MA17	Prulifloxacin	5
J01MA18	Pazufloxacin	5
J01MA19	Garenoxacin	5
J01MA21	Sitafoxacin	5
J01MA22	Tosufloxacin	5
J01MA23	Delafloxacin	5
J01MB	Other Quinolones	4
J01MB01	Rosoxacin	5
J01MB02	Nalidixic Acid	5
J01MB03	Piromidic Acid	5
J01MB04	Pipemidic Acid	5
J01MB05	Oxolinic Acid	5
J01MB06	Cinoxacin	5
J01MB07	Flumequine	5
J01MB08	Nemonoxacin	5
J01R	Combinations Of Antibacterials	3
J01RA	Combinations Of Antibacterials	4
J01RA01	Penicillins, Combinations With Other Antibacterials	5
J01RA02	Sulfonamides, Combinations With Other Antibacterials (Excl. Trimethoprim)	5
J01RA03	Cefuroxime And Metronidazole	5

J01RA04	Spiramycin And Metronidazole	5
J01RA05	Levofloxacin And Ornidazole	5
J01RA06	Cefepime And Amikacin	5
J01RA07	Azithromycin, Fluconazole And Secnidazole	5
J01RA08	Tetracycline And Oleandomycin	5
J01RA09	Ofloxacin And Ornidazole	5
J01RA10	Ciprofloxacin And Metronidazole	5
J01RA11	Ciprofloxacin And Tinidazole	5
J01RA12	Ciprofloxacin And Ornidazole	5
J01RA13	Norfloxacin And Tinidazole	5
J01X	Other Antibacterials	3
J01XA	Glycopeptide Antibacterials	4
J01XA01	Vancomycin	5
J01XA02	Teicoplanin	5
J01XA03	Telavancin	5
J01XA04	Dalbavancin	5
J01XA05	Oritavancin	5
J01XB	Polymyxins	4
J01XB01	Colistin	5
J01XB02	Polymyxin B	5
J01XC	Steroid Antibacterials	4
J01XC01	Fusidic Acid	5
J01XD	Imidazole Derivatives	4
J01XD01	Metronidazole	5
J01XD02	Tinidazole	5
J01XD03	Ornidazole	5
J01XE	Nitrofurans Derivatives	4
J01XE01	Nitrofurantoin	5
J01XE02	Nifurtimol	5
J01XE03	Furazidin	5
J01XE51	Nitrofurantoin, Combinations	5
J01XX	Other Antibacterials	4
J01XX01	Fosfomycin	5
J01XX02	Xibornol	5
J01XX03	Clofoctol	5
J01XX04	Spectinomycin	5
J01XX05	Methenamine	5
J01XX06	Mandelic Acid	5
J01XX07	Nitroxoline	5
J01XX08	Linezolid	5
J01XX09	Daptomycin	5
J01XX10	Bacitracin	5
J01XX11	Tedizolid	5

Annex 5: WHO AWARE Categories for Antimicrobials

The 2021 edition of the WHO Model List of Essential Medicines introduced a classification of antibiotics into three classes, with examples in the table below:

- **Access Antibiotics:** first- or second-line medicines for most common clinical infective syndromes, based on comprehensive review of 21 priority infectious syndromes in children and adults, review of 5 bacterial infections in children (community acquired pneumonia, dysentery, cholera, sepsis, severe acute malnutrition), and review for antibiotics for STI (current WHO guidelines). These antibiotics should be widely available, accessible and quality assured.
- **Watch Antibiotics:** classes of antibiotics with a broad spectrum and a higher risk of resistance, to be used for selected indications and to be considered as a focus for stewardship. This group includes most of the highest priority agents among the Critically Important Antimicrobials for Human Medicine. Some ACCESS antibiotics, like ceftriaxone and azithromycin are also in the WATCH group.
- **Reserve Antibiotics:** “last resort” antibiotics, which should be restricted only for multi resistant infections and for targeted treatment and should also be focus for stewardship.

AWaRe Category	Antibiotic	Antibiotic Class
Access	Amikacin	Aminoglycosides
	Amoxicillin	Penicillins
	Amoxicillin/clavulanic-acid	Beta-lactam/beta-lactamase-inhibitor
	Ampicillin	Penicillins
	Ampicillin/sulbactam	Beta-lactam/beta-lactamase-inhibitor
	Azidocillin	Penicillins
	Bacampicillin	Penicillins
	Benzathine-benzylpenicillin	Penicillins
	Benzylpenicillin	Penicillins
	Brodinoprim	Trimethoprim-derivatives
	Cefacetrile	First-generation-cephalosporins
	Cefadroxil	First-generation-cephalosporins
	Cefalexin	First-generation-cephalosporins
	Cefaloridine	First-generation-cephalosporins
	Cefalotin	First-generation-cephalosporins
	Cefapirin	First-generation-cephalosporins
	Cefatrizine	First-generation-cephalosporins
	Cefazedone	First-generation-cephalosporins
	Cefazolin	First-generation-cephalosporins
	Cefradine	First-generation-cephalosporins
	Cefroxadine	First-generation-cephalosporins
	Ceftazole	First-generation-cephalosporins
	Chloramphenicol	Amphenicols
	Clindamycin	Lincosamides
	Clometocillin	Penicillins
	Cloxacillin	Penicillins
	Dicloxacillin	Penicillins
	Doxycycline	Tetracyclines
	Epicillin	Penicillins
	Flucloxacillin	Penicillins

	Furazidin	Nitrofuran derivatives
	Gentamicin	Aminoglycosides
	Hetacillin	Penicillins
	Mecillinam	Penicillins
	Metampicillin	Penicillins
	Meticillin	Penicillins
	Metronidazole_IV	Imidazoles
	Metronidazole_oral	Imidazoles
	Nafcillin	Penicillins
	Nifurtinol	Nitrofuran derivatives
	Nitrofurantoin	Nitrofuran-derivatives
	Ornidazole_IV	Imidazoles
	Ornidazole_oral	Imidazoles
	Oxacillin	Penicillins
	Penamecillin	Penicillins
	Phenoxymethylpenicillin	Penicillins
	Pivampicillin	Penicillins
	Pivmecillinam	Penicillins
	Procaine-benzylpenicillin	Penicillins
	Propicillin	Penicillins
	Secnidazole	Imidazoles
	Spectinomycin	Aminocyclitols
	Sulbactam	Beta-lactamase-inhibitors
	Sulfadiazine	Sulfonamides
	Sulfadiazine/tetroxoprim	Sulfonamide-trimethoprim-combinations
	Sulfadiazine/trimethoprim	Sulfonamide-trimethoprim-combinations
	Sulfadimethoxine	Sulfonamides
	Sulfadimidine	Sulfonamides
	Sulfadimidine/trimethoprim	Sulfonamide-trimethoprim-combinations
	Sulfafurazole	Sulfonamides
	Sulfaisodimidine	Sulfonamides
	Sulfalene	Sulfonamides
	Sulfamazone	Sulfonamides
	Sulfamerazine	Sulfonamides
	Sulfamerazine/trimethoprim	Sulfonamide-trimethoprim-combinations
	Sulfamethizole	Sulfonamides
	Sulfamethoxazole	Sulfonamides
	Sulfamethoxazole/trimethoprim	Sulfonamide-trimethoprim-combinations
	Sulfamethoxypyridazine	Sulfonamides
	Sulfametomidine	Sulfonamides
	Sulfametoxydiazine	Sulfonamides
	Sulfametrole/trimethoprim	Sulfonamide-trimethoprim-combinations
	Sulfamoxole	Sulfonamides
	Sulfamoxole/trimethoprim	Sulfonamide-trimethoprim-combinations
	Sulfanilamide	Sulfonamides
	Sulfaperin	Sulfonamides
	Sulfaphenazole	Sulfonamides
	Sulfapyridine	Sulfonamides
	Sulfathiazole	Sulfonamides
	Sulfathiourea	Sulfonamides

	Sultamicillin	Beta-lactam/beta-lactamase-inhibitor
	Talampicillin	Penicillins
	Tetracycline	Tetracyclines
	Thiamphenicol	Amphenicols
	Tinidazole_IV	Imidazoles
	Tinidazole_oral	Imidazoles
	Trimethoprim	Trimethoprim-derivatives
Watch	Arbekacin	Aminoglycosides
	Arbekacin	Aminoglycosides
	Aspoxicillin	Penicillins
	Azithromycin	Macrolides
	Azlocillin	Penicillins
	Bekanamycin	Aminoglycosides
	Biapenem	Carbapenems
	Carbenicillin	Penicillins
	Carindacillin	Penicillins
	Cefaclor	Second-generation-cephalosporins
	Cefamandole	Second-generation-cephalosporins
	Cefbuparazone	Second-generation-cephalosporins
	Cefcapene-pivoxil	Third-generation-cephalosporins
	Cefdinir	Third-generation-cephalosporins
	Cefditoren-pivoxil	Third-generation-cephalosporins
	Cefepime	Fourth-generation-cephalosporins
	Cefetamet-pivoxil	Third-generation-cephalosporins
	Cefixime	Third-generation-cephalosporins
	Cefmenoxime	Third-generation-cephalosporins
	Cefmetazole	Second-generation-cephalosporins
	Cefminox	Second-generation-cephalosporins
	Cefodizime	Third-generation-cephalosporins
	Cefonicid	Second-generation-cephalosporins
	Cefoperazone	Third-generation-cephalosporins
	Ceforanide	Second-generation-cephalosporins
	Cefoselis	Fourth-generation-cephalosporins
	Cefotaxime	Third-generation-cephalosporins
	Cefotetan	Second-generation-cephalosporins
	Cefotiam	Second-generation-cephalosporins
	Cefoxitin	Second-generation-cephalosporins
	Cefozopran	Fourth-generation-cephalosporins
	Cefpiramide	Third-generation-cephalosporins
	Cefpirome	Fourth-generation-cephalosporins
	Cefpodoxime-proxetil	Third-generation-cephalosporins
	Cefprozil	Second-generation-cephalosporins
	Cefsulodin	Third-generation-cephalosporins
	Ceftazidime	Third-generation-cephalosporins
	Cefteram-pivoxil	Third-generation-cephalosporins
	Ceftibuten	Third-generation-cephalosporins
	Ceftizoxime	Third-generation-cephalosporins
	Ceftriaxone	Third-generation-cephalosporins
	Cefuroxime	Second-generation-cephalosporins
	Chlortetracycline	Tetracyclines

Cinoxacin	Quinolones
Ciprofloxacin	Fluoroquinolones
Clarithromycin	Macrolides
Clofoctol	Phenol derivatives
Clomocycline	Tetracyclines
Delafloxacin	Fluoroquinolones
Demeclocycline	Tetracyclines
Dibekacin	Aminoglycosides
Dirithromycin	Macrolides
Doripenem	Carbapenems
Enoxacin	Fluoroquinolones
Ertapenem	Carbapenems
Erythromycin	Macrolides
Fidaxomicin	Macrolides
Fleroxacin	Fluoroquinolones
Flomoxef	Second-generation-cephalosporins
Flumequine	Quinolones
Flurithromycin	Macrolides
Fosfomycin_oral	Phosphonics
Fusidic-acid	Steroid antibacterials
Garenoxacin	Fluoroquinolones
Gatifloxacin	Fluoroquinolones
Gemifloxacin	Fluoroquinolones
Grepafloxacin	Fluoroquinolones
Imipenem/cilastatin	Carbapenems
Isepamicin	Aminoglycosides
Josamycin	Macrolides
Kanamycin_IV	Aminoglycosides
Kanamycin_oral	Aminoglycosides
Lascufloxacin	Fluoroquinolones
Latamoxef	Third-generation-cephalosporins
Levofloxacin	Fluoroquinolones
Levonadifloxacin	Fluoroquinolones
Lincomycin	Lincosamides
Lomefloxacin	Fluoroquinolones
Loracarbef	Second-generation-cephalosporins
Lymecycline	Tetracyclines
Meropenem	Carbapenems
Metacycline	Tetracyclines
Mezlocillin	Penicillins
Micronomicin	Aminoglycosides
Midecamycin	Macrolides
Minocycline_oral	Tetracyclines
Miocamycin	Macrolides
Moxifloxacin	Fluoroquinolones
Nemonoxacin	Quinolones
Neomycin_IV	Aminoglycosides
Neomycin_oral	Aminoglycosides
Netilmicin	Aminoglycosides
Norfloxacin	Fluoroquinolones
Ofloxacin	Fluoroquinolones

	Oleandomycin	Macrolides
	Oxolinic-acid	Quinolones
	Oxytetracycline	Tetracyclines
	Panipenem	Carbapenems
	Pazufloxacin	Fluoroquinolones
	Pefloxacin	Fluoroquinolones
	Penimepicycline	Tetracyclines
	Pheneticillin	Penicillins
	Pipemidic-acid	Quinolones
	Piperacillin	Penicillins
	Piperacillin/tazobactam	Beta-lactam/beta-lactamase-inhibitor_anti-pseudomonal
	Piromidic-acid	Quinolones
	Pristinamycin	Streptogramins
	Prulifloxacin	Fluoroquinolones
	Ribostamycin	Aminoglycosides
	Rifabutin	Rifamycins
	Rifampicin	Rifamycins
	Rifamycin_IV	Rifamycins
	Rifamycin_oral	Rifamycins
	Rifaximin	Rifamycins
	Rokitamycin	Macrolides
	Rolitetracycline	Tetracyclines
	Rosoxacin	Quinolones
	Roxithromycin	Macrolides
	Rufloxacin	Fluoroquinolones
	Sarecycline	Tetracyclines
	Sisomicin	Aminoglycosides
	Sitafloxacin	Fluoroquinolones
	Solithromycin	Macrolides
	Sparfloxacin	Fluoroquinolones
	Spiramycin	Macrolides
	Streptoduocin	Aminoglycosides
	Streptomycin_IV	Aminoglycosides
	Streptomycin_oral	Aminoglycosides
	Sulbenicillin	Penicillins
	Tazobactam	Beta-lactamase-inhibitors
	Tebipenem	Carbapenems
	Teicoplanin	Glycopeptides
	Telithromycin	Macrolides
	Temafloxacin	Fluoroquinolones
	Temocillin	Penicillins
	Ticarcillin	Penicillins
	Tobramycin	Aminoglycosides
	Tosufloxacin	Fluoroquinolones
	Troleandomycin	Macrolides
	Trovafoxacin	Fluoroquinolones
	Vancomycin_IV	Glycopeptides
	Vancomycin_oral	Glycopeptides
Reserve	Aztreonam	Monobactams
	Carumonam	Monobactams

Annex 6: Tool for WHO Indicator surveys

How to Investigate Drug Use in Health Facilities: Selected Drug Use Indicators

https://www.who.int/medicines/publications/how-to-investigate_drug-use/en/

Annex 7: Instructions for Drug Indicator Surveys

Practical instructions for DIS

Study Setting:

These surveys will be conducted in the general OPD, excluding antenatal and specialist out-patient clinics such as diabetes, HIV, Neurology and mental health. Prescription practices in these departments will be very different.

Inclusion/Exclusion criteria:

In order to have a single sample for the DIS and more specific Antibiotic survey, the inclusion criteria will be ALL patients seen at general OPD regardless of whether they were prescribed an antibiotic or not.

OPD re-attendances and visits that result in admissions are excluded.

Sample size:

The suggested sample size is at least 200 patients/prescriptions within the period of consideration. Considering the high rate of antibiotic use in primary health care settings, this will produce a sample of more than 100 patients for which antibiotics were prescribed.

Sampling strategy

The suggested sampling strategy is systematic sampling. For the period considered (2-3 months), the total number of entries in the OPD Register, divided by the expected sample size will provide the sampling interval.

- From HMIS 002, get the number of OPD (new) visits for the 3 months you have decided to investigate: e.g. 3456 new OPD visits in the months January to March 2016.
- Divide the total number by the number of prescriptions you want to sample and round the result: e.g. $3456/100 = 34.56$ rounded down to 34.
- Choose a random number from 1 to 9 (common method is to take out a banknote and take the last figure of the serial number) and sample one patient every 35 (in this example) starting from the patient number indicated by the random number.
- Skip re-attendances and admissions while counting.
- Decide what to do in case the prescription does not fit the definition (e.g. if it is an admission case), i.e., choose the previous prescription or the next.

Data collection:

The data source is the OPD Register (HMIS 002). The variables to be collected include:

- Serial Number/OPD Number
- Age
- Sex
- weight
- Test done*
- Diagnosis
- Drugs prescribed including dose and duration

*Data on these variables may not be captured from the OPD Register.

Collect and analyse the data using the attached form and formulas (see Table 9 below). There are no pre-set absolute thresholds or standards for the value of the indicators, since they depend on a number of factors. Further analysis will be carried out on this data set, therefore, the most practical approach is to copy the complete prescription of the sampled patients and complete the indicator table.

Table 9: Formulas for the indicator survey

Patient Number	No. of medicines prescribed	No. of medicines prescribed by generic name	No. of antibiotics prescribed	No. of injections prescribed	Number of medicines not in the UCG/EMHSL	Diagnosis recorded Y/N
1	3	3	1	0	3	Y
2	2	2	1	0	2	Y
3	3	3	2	0	3	Y
4	3	2	0	0	2	Y
5						
6						
Total no. medicines	A	B	C	D	E	
Total no. patients	F		G	H		I
Indicator	AVERAGE NUMBER OF MEDICINES PER PATIENT (Total meds /#patients) = A/F	% OF MEDICINES PRESCRIBED BY GENERIC NAME = B/A x 100	% PATIENTS RECEIVING 1 OR MORE ANTIBIOTICS = G/F x 100 %of medicines being antibiotics = C/A x 100	% PATIENTS RECEIVING 1 OR MORE INJECTIONS = H/F x 100 %of medicines being injections = D/A x 100	% OF MEDICINES NOT IN THE UCG = E/A x 100	% DIAGNOSIS RECORDING = I/F x 100

Table 10: OPD data collection tool

Region:															
District:															
Facility Name:															
Level of Care															
Ownership: (Public, PNFP, PFP)															
Date of Audit:															
Data sources		OPD Register												Lab Register	
No	Prescription date	Patient Number	Patient Initials	Age	Sex	Diagnosis	Medicines prescribed	Medicine is an Antibiotic? (Y/N)	Oral OR Inject-able	Medicines prescribed by Generic name (Y/N)	Adherence to treatment guideline (Y/N)	Treatment based on biomarker (Y/N)	biomarker? (WBC count, PCT, CRP)	Type of biological sample (urine, blood, etc)	

Annex 8: Instructions for Medicines Use Evaluation and Prescription Audit

Practical instructions for MUE and prescription audit

Step 1: Identify a priority condition or antimicrobial of interest. (it can be a diagnosis e.g. sepsis, pneumonia, or a medicine e.g. an expensive antibiotic). Define the scope of the activity, which refers to the parameters you are going to assess, i.e. prescribing criteria, dispensing, and administration. The choice depends on the problem you are looking at.

For example, if the problem pointer is a high number of treatment failures in patients using an antibiotic, you may want to investigate indication but also dosages, the way it is administered or prepared, patient characteristics and culture and sensitivity results.

Step 2: Detail the standard management criteria according to guidelines (IMCI, UCG, PGD). To keep it simple, limit to 3–5 criteria. Create a simple data collection tool based on the established criteria.

Step 3: Set the threshold below which the adherence to standard would be considered insufficient: often 100% is unrealistic, 90%–95% is sufficient in most cases.

Step 4: Describe how the data will be collected. This is an important consideration because while some data are easy to collect retrospectively, some others can only be collected prospectively.

Step 5: Establish the number of prescriptions to be analysed: minimum 30, but up to 100 for common conditions/medicines, and in big facilities with multiple prescribers.

Step 6: For retrospective studies

For a prescription audit, establish the period you want to investigate (usually 1–3 months). Obtain the total number of cases with the condition under investigation from the HMIS for that period and divide it by the number of prescriptions you want to collect: the result will be your sampling interval.

Example: if you are doing a prescription audit on Urinary Tract Infection (UTI), and you want 50 prescriptions from a period of 1 month: check how many UTI cases are recorded in HMIS 105 for that month (e.g. 346) and divide by 50. That is, $346/50 = 7$, so you will record every 7th case of UTI from the OPD register.

For a medicine use evaluation, establish the period you want to investigate, check how many patients have been prescribed the medicine in the period of interest, divide it by the number of prescriptions you want to collect and use the result for your sampling interval.

Example: you want to do a prescription survey on ceftriaxone. You may get the number of patients dispensed ceftriaxone in a certain period from the pharmacy dispensing log, e.g. 155. Divide the number by the number of prescriptions you are targeting (30) to obtain your sampling interval. That is, $155/30 = 5$, so you will every 5th patient prescribed metformin from the OPD register.

Step 6: For prospective studies. These are often based on observation, and the sample size may depend on the amount of time available, and the number of cases per day. Usually when health workers are aware that they are being observed, they may change their behaviour but they soon get used and revert to usual practices. So it is advisable to start collecting data after having done some observations. Prospective methods have risks of bias since data are collected in a short period and there is a limited chance of random sampling, so they are used in case of absence of retrospective data (poor records) or to study certain practices (e.g. how nurses prepare and administer injectable medicines).

Step 7: Collect data (retrospectively or prospectively) and tabulate them for analysis. If documentation is poor, the only way to collect data is prospectively. Analyse percentage of adherence to criteria and compile a report with recommendations.

Step 8: if the problem has a straightforward solution, share the report with prescribers, then design and implement an intervention. If the reasons of the problem have to be investigated, design and conduct qualitative studies to inform the development of the intervention.

Step 9: Repeat the medicine use evaluation or the prescription audit during and after the intervention for monitoring and evaluation purposes. Remember that data collection, analysis and feedback to prescribers by itself it is an intervention because it can influence prescribers' behaviour.

Annex 9: Template for Point Prevalence Surveys

Patient form: Point Prevalence Survey on Antibiotic Use

CORE: Patient demographics

Hospital ID		Hospital Code	
Ward ID		Ward Code	
Survey Date		Patient Code	
Patient Initials		AgeYear (>2 years)	
Gender		PreTermBirth	
Age (0-23 months)		NeonatesBirthWeight	
Child12YearWeight		SurgerySinceAdmission	
AdmissionDate		PeripheralVascularCatheter	
CentralVascularCatheter		Intubation	
UrinaryCatheter			
PatientOnAntibiotic		PatientNumberAntibiotics	

OPTIONAL: Underlying infectious diseases variable

MalariaStatus		TuberculosisStatus	
HIVStatus		HIVOnART*	
HIVCD4Count*		<i>*Only if HIVStatus = "Yes"</i>	

OPTIONAL: Comorbidities variables

COPDStatus		MalnutritionStatus	
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OPTIONAL: Hospitalisation variables

TransferfromHospital		TransferfromNonHospital	
Hospitalisation90Days			

OPTIONAL: Hospitalisation variables

TypeSurgerySinceAdmission	
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Indication form: Point Prevalence Survey on Antibiotic Use

Core variables

Patient Code Ward Code Hospital Code

IndicationCounter	1	2	3	4
IndicationType	HAI/CAI/SP/MP/O	HAI/CAI/SP/MP/O	HAI/CAI/SP/MP/O	HAI/CAI/SP/MP/O
If IndicationType is SP:				
Surg.Proph.Duration	SP1/SP2/SP3	SP1/SP2/SP3	SP1/SP2/SP3	SP1/SP2/SP3
Surg.Proph.Site				
Diagnosis				
StartDateTreatment				
ReasonInNotes	YES/NO	YES/NO	YES/NO	YES/NO
CultureSampleTaken	YES/NO/UNK	YES/NO/UNK	YES/NO/UNK	YES/NO/UNK

IndicationType: HAI = Hospital Acquired Infection, CAI = Community Acquired Infection, SP = Surgical Prophylaxis, MP = Medical Prophylaxis, O = Other
Surg.Proph.Duration: SP1 = one dose, SP2 = Multiple doses on one day (up to 24 hours after the surgery), SP3 = multiple doses for more than 24 hours after the surgery

Antibiotic form (1): Point Prevalence Survey on Antibiotic Use

Patient Code Ward Code Hospital Code

Core variables

Antibiotic counter	Indication Counter	Antibiotics Notes Name	Antibiotic INN Name	Antibiotic Written in INN	Start Date of Antibiotic	Unit Dose	Unit Dose Combination	Unit Dose Measure/Unit	Unit Dose Frequency	Administration Route
1				YES NO				MG G IU MU		O P I R
2				YES NO				MG G IU MU		O P I R
3				YES NO				MG G IU MU		O P I R
4				YES NO				MG G IU MU		O P I R
5				YES NO				MG G IU MU		O P I R
6				YES NO				MG G IU MU		O P I R

Administration Route: O = Oral, P = Parenteral, I = Inhalation, R = Rectal

Antibiotic form (2): Point Prevalence Survey on Antibiotic Use

Patient Code
Ward Code
Hospital Code

Core variables

Antibiotic counter	Prescriber Type	Parenteral Type	Oral Switch	Number of Missed doses	Reason for Missed doses	Guideline compliance	Treatment type
1	SP GP O N	IM IV O	YES NO		S P O UNK	Yes No Not indicated	MG G IU MU
2	SP GP O N	IM IV O	YES NO		S P O UNK	Yes No Not indicated	MG G IU MU
3	SP GP O N	IM IV O	YES NO		S P O UNK	Yes No Not indicated	MG G IU MU
4	SP GP O N	IM IV O	YES NO		S P O UNK	Yes No Not indicated	MG G IU MU
5	SP GP O N	IM IV O	YES NO		S P O UNK	Yes No Not indicated	MG G IU MU
6	SP GP O N	IM IV O	YES NO		S P O UNK	Yes No Not indicated	MG G IU MU

Prescriber type: SP = Specialist, GP = General Practitioner, O = Other, N = Unknown

Parenteral type: IM = Intramuscular, IV = Intravenous, O = Other

Reason for Missed doses: S = Stock out, P = Failure to Purchase, O = Other, UNK = Unknown

Microbiology form: Point Prevalence Survey on Antibiotic Use

Hospital Code

Ward Code

Patient Code

Microbiology data refers to any culture & susceptibility result from a relevant clinical sample. Screening samples should not be reported

Specimen 1:

Specimen Type

Culture Result

NA/POS/NEG

If culture result is Pos:

Microorganism

1

2

3

Antibiotic Susceptibility Test Results

1

2

3

(Only if Antibiotic Susceptibility Test Result is "Yes")

1

2

3

Specimen 2:

Specimen Type

Culture Result

NA/POS/NEG

If culture result is Pos:

Microorganism

1

2

3

Antibiotic Susceptibility Test Results

1

2

3

1

2

3

Specimen 3:

Specimen Type

Culture Result

NA/POS/NEG

If culture result is Pos:

Microorganism

1

2

3

Antibiotic Susceptibility Test Results

1

2

3

1

2

3

Culture result: **NA** = Not available. **POS** = Positive, **NEG** = Negative

Annex 10: Attendance Stakeholder Engagement

Sl. No.	First Name	Surname	Cadre	Institution
1	Winnie	Nambatya	Lecturer	Makerere University Department of Pharmacy
2	Kalidi	Rajab	Lecturer	Makerere University Department of Pharmacy
3	Bruhan	Kaggwa	Lecturer	Makerere University Department of Pharmacy
4	Edson	Munanura	Lecturer	Makerere University Department of Pharmacy
5	Rodney	Tabaruka	Principal. Pharmacist	Ministry of Health
6	Phillip	Ampaire	Senior Pharmacist	Ministry of Health
7	Daniel	Aguma	Senior Pharmacist	Ministry of Health
8	Peter	Agababingi	Monitoring and Evaluation Officer	Ministry of Health
9	Martha Grace	Ajulong	Ag Commissioner Health Services - Pharmaceuticals and Natural Medicine	Ministry of Health
10	Victor	Bewayo	Pharmacist	Arua Regional Referral Hospital
11	Chrispus	Ngabirano	Microbiologist	Kabale Regional Referral Hospital
12	Harriet	Akelo	Senior Pharmacist	Ministry of Health
13	Micheal	Isabirye	Capacity Building Coordinator	Ministry of Health
14	Christopher	Amandu Harold	Senior Laboratory Technologist	Arua Regional Referral Hospital
15	James	Achol	Pharmacist	Jinja Regional Referral Hospital
16	Enock	Padere	HODM	Iganga General Hospital
17	Falisy	Lule	Senior Pharmacist	Kawempe Regional Referral Hospital
18	Joanitah	Atuhaire	Regulatory officer	National Drug Authority
19	Prossy	Atimango	Public Health Officer	Ministry of Health
20	William	Olum Pjathim	Senior Pharmacist	Jinja Regional Referral Hospital
21	Ventrine	Chelimo	Pharmacit	National Medical Stores
22	Harriet	Tino Okello	Pharmacist	Lira Regional Referral Hospital
23	Morries	Seru	Rtd. Commissioner Health Services - DPNM	Ministry of Health
24	David	Arinaitwe	Senior Pharmacist	National Medical Stores
25	Isaac	Mukama	Software Developer	Jhpiego
26	Sandra	Namakula	Data Manger	Infectious Disease Institute
27	Christopher	Wagobera	Medical Officer	Kabale Regional Referral Hospital
28	Paul	Rubayinza	Lecturer	Makerere University Department of Pharmacy
29	Theophile	Tuyishimire	Pharmacist	National Medical Stores
30	Jaqueline	Nassuna	Pharmacist	National Drug Authority
31	Calvin	Chemutai	Data Analyst	Infectious Disease Institute
32	Emmanuel	Nkurunziza	Data Officer	Infectious Disease Institute
33	Hanifa	Nakwenda	GIS specialist	Infectious Disease Institute
34	Vivian	Twemanye	Senior Project Officer - Antimicrobial Consumption and Use	Infectious Disease Institute
35	Sr Josephine	Oyella	Senior Pharmacist	St Mary's Hospital Lacor
36	Denis	Nankoola	Senior Pharmacist	Fort Portal Regional Referral Hospital

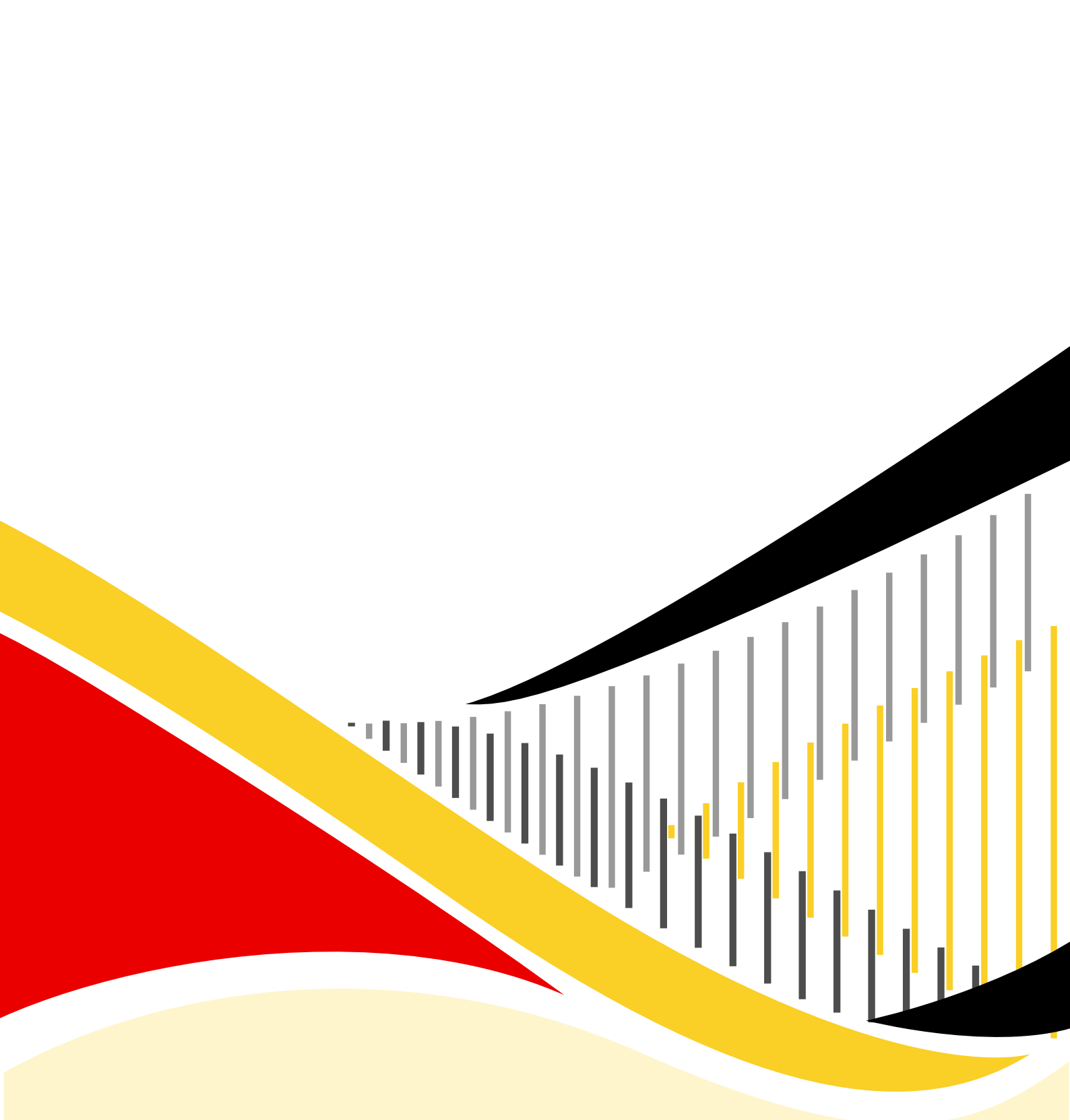
37	Zainab	Akello	Pharmacist	Gulu Regional Referral Hospital
38	Rogers	Kisame	Program Manger	Baylor Foundation Uganda
39	Moses	Mukiibi	Antimicrobial Use and Consumption Coordinator	Baylor Foundation Uganda
40	Sheila	Ampaire	Regulatory officer	National Drug Authority
41	Jakira	Ambrose	Health Product Quantification Specilist	Ministry of Health
42	Ian	Nyamitoro	Senior Health Product Management officer	Ministry of Health
43	Thomas	Ssemakadde	Laboratory Systems Coordinator	Baylor Foundation Uganda
44	Reginald Rony	Bahatungire	Ag Commissioner Health Services - Clinical Services	Ministry of Health
45	Joel Tutu	Miti	Pharmacist	Ministry of Health
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64	Patrick	Opio	Senior Pharmacist	Mubende Regional Referral Hospital
65	George	Katongole	Laboratory Technologist	Mubende Regional Referral Hospital
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67	Helen	Kabagambe	Senior Pharmacist	Naguru Regional Referral Hospital
68	Joseph	Olore	Senior Laboratory Technician	Naguru Regional Referral Hospital
69	William	Oyang	Consultant Padiatrician	Lira Regional Referral Hospital
70	Benard	Ongora	Laboratory Technologist	Lira Regional Referral Hospital
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74	Sophie	Nakitto	Pediatrician	Kayunga Regional Referral Hospital
75	Daniel	Kibombo	Microbiologist	Kayunga Regional Referral Hospital

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79	Wilson	Etolu	Consultant Physician	Soroti Regional Referral Hospital
80	Anthony	Makhoba	Consultant Physician	Fortportal Regional Referral Hospital
81	Simon Peter	Seguya	Senior Pharmacist	Fortportal Regional Referral Hospital
82	Dan	Kakyakumaiso	Senior Medical Laboratory Technologist	Fortportal Regional Referral Hospital
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86	Enoch	Padere	Laboratory Technologist	Jinja Regional Referral Hospital
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88	Stephen	Odomel	Pharmacist	Moroto Regional Referral Hospital
89	Bosco	Adranya	Senior Laboratory Technologist	Moroto Regional Referral Hospital
90	Alex	Sande	Senior Pharmacist	Mbale Regional Referral Hospital
91	Asad	Muyinda	Consultant Internal Medicine	Mbale Regional Referral Hospital
92	Dauson	Wanyibe	Senior Laboratory Technician	Mbale Regional Referral Hospital
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95	Francis	Oriokot	Senior Consultant	Mbarara Regional Referral Hospital
96	Robert	Wagubi	Laboratory Technician	Mbarara Regional Referral Hospital
97	Tuhaise	Gamukama	Surgeon	Kabale Regional Referral Hospital
98	Patrick	Odong Olwedo	Executive Consultant	Yumbe Regional Referral Hospital
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